DSO203 / HW 2.81 - 2.82

WILDCAT

User Manual 6.6

Oscilloscope / 144 MS/s / FFT / Generator / Decoder

WHAT IS NEW

144 MS/s interleaved CH A

FPGA W1.2; corrected two-ADC sampling, fast timebases

Unified calibration

Normal A/B + 144 B-to-A ZERO and DC REF

Verified calibration storage

Dual WPT/WPB generations and power profiles

Based on the Wildcat 6.5b illustrated manual (41 pages)

Technical verification: repository bales0/DSO203-HW-2.81, commit 4df9eba

22 October 2026

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Updated illustrated operating guide for Wildcat 6.6 on DS203 HW 2.81/2.82.

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How to use this edition: The illustrated Normal-mode chapters are retained from the older reference. The new Section 04A covers the implemented 144 MS/s mode. The rewritten Sections 10 and 11 replace outdated 6.5b calibration and compatibility instructions. The release notes distinguish implemented features from planned work.

Complete illustrated reference / 49 pages / code state: 4df9eba

SECTION 01

Introduction, controls and flash layout

Wildcat 6.6 for DS203 HW 2.81/2.82 - hardware orientation, first measurement and firmware allocation.

The DSO203 is a compact four-channel instrument with two analogue inputs, two digital inputs and a waveform-generator output. In Wildcat 6.5b the top buttons are context-sensitive; their action depends on whether the key is pressed briefly, held or used while a particular menu is active.

The drawing below follows the approved layout: original labels and surrounding border have been removed, and six numbered markers point to the actual top controls. The other markers identify physical connectors. Functions are explained in the accompanying reference table.

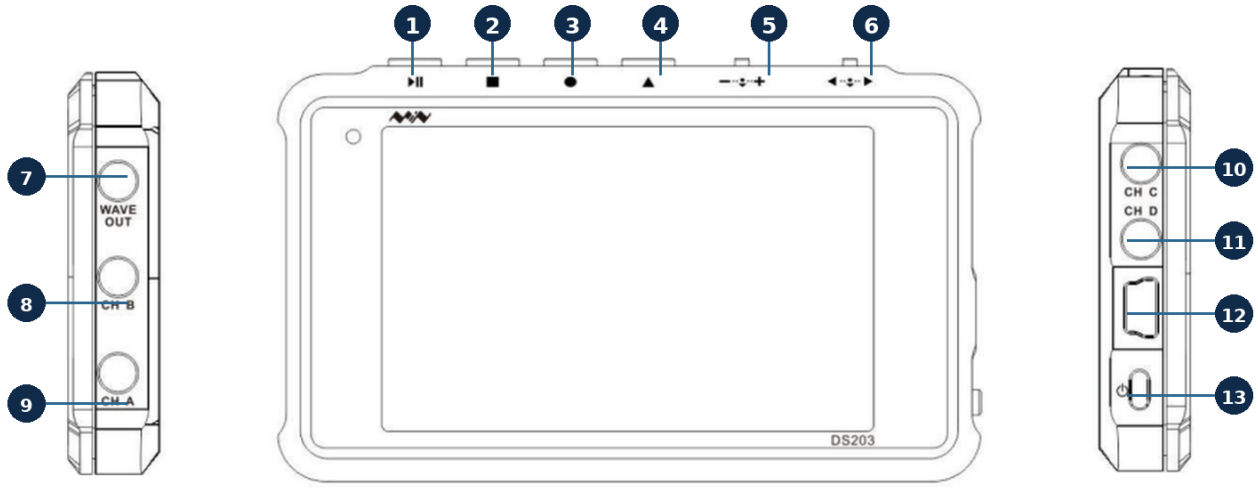


FIG. 01 DSO203 front and side views, with numbered controls and connectors.

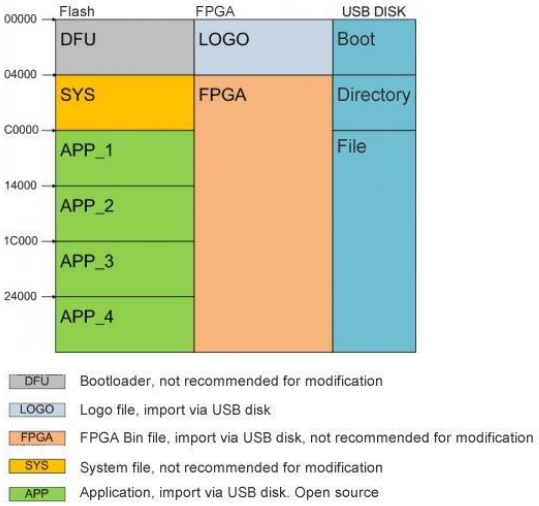
No.	Wildcat control	Function
1	Button 1 — RUN/HOLD	Short press toggles acquisition. Long press opens V/div shortcut; double-tap engages AUTO-SET.
2	Button 2 — Trigger / calibration	Short press selects automatic trigger level; long press selects manual level. Hold for calibration or a contextual adjustment.
3	Button 3 — Meters / configuration	Short press cycles meter layouts; long press enters standby; hold to save configuration.
4	Button 4 — Menu item	Short press selects the next item; long press jumps to time/div; hold toggles cursor-restricted meters.
5	Left toggle	Left/right adjusts the selected value; centre press controls trigger source or context-sensitive functions.
6	Right toggle	Left/right selects the menu; centre press selects acquisition buffer mode or another context-sensitive function.
7–9	Output / CH B / CH A	Waveform generator output, analogue B input and analogue A input.
10–13	CH C / CH D / USB / Power	Two digital inputs, USB connection and the power switch.

First measurement

- Connect the signal to CH A. Set an appropriate volts-per-division range and choose AC or DC coupling as needed.
- Long-press Button 4 to jump to TIME/DIV and adjust the timebase to display a complete signal period.
- If the waveform is hard to identify, double-tap Button 1 for AUTO-SET. Press Button 1 once to stop or resume acquisition.
- Short-press Button 3 to cycle regular, large or hidden meters.

Internal flash and application-bank allocation

The historical DS203 memory chart describes the four application slots, system region and FPGA storage separately. Wildcat 6.5b extends across APP_1 and APP_2 and uses two additional application-code regions. APP_3 and APP_4 are not allocated by the Wildcat linker and remain free for user applications.



Original DS203 memory layout (schematic, not to scale for the later HW 2.81 / 2.82).

From	Through	Bank / area	Use
0x08000000	0x08003FFF	DFU	Bootloader — not Wildcat
0x08004000	0x0800BFFF	SYS	System firmware — separate from Wildcat
0x0800C000	0x08013FFF	APP_1	Wildcat main ROM, first 32 KiB
0x08014000	0x0801BFFF	APP_2	Wildcat main ROM, second 32 KiB
0x0801C000	0x08023FFF	APP_3	Free for user application
0x08024000	0x0802BFFF	APP_4	Free for user application
0x0802C000	0x0803FFFF	Extended 1	Wildcat extra code; 80 KiB linker region
0x08040000	0x08046FFF	Reserved	Outside Wildcat linker allocations
0x08047000	0x080573FF	Extended 2	Wildcat extra code; 65 KiB linker region
0x08057400	0x0805C7FF	Reserved	Between extra code and FPGA

From	Through	Bank / area	Use
0x0805C800	0x0807D7B7	FPGA W1.2	135,096-byte bitstream

The Wildcat code ranges show linker allocation limits, not the number of programmed bytes. The FPGA row gives the bitstream length. Do not overwrite DFU, SYS, extended code or FPGA regions as though they were normal application slots.

WILDCAT 6.6 FLASH NOTE

The FPGA W1.2 bitstream occupies the same protected FPGA region. 144 MS/s entry requires verified W1.2, while legacy Normal functions remain documented in preceding sections. Never flash a single flat BIN across the disjoint application regions.

SECTION 02

Normal mode capabilities

A guided overview of the enhancements over earlier stock and community builds.

SUPPORT FOR 8MB DEVICES AND HARDWARE VERSION 2.81

RELIABLE TRIGGERING: Efforts have been made to improve triggering performance, something that was notably poor with early programs shipped with the units. Note however that some time base modes have inherent overhead that can cause skipped triggers with random signals.

For example, the AUTO mode times out waiting for triggers after a set time so it can auto-scan. While auto-scanning, the frame must finish before any trigger is detected. Most reliable triggering on random or intermittent signals is achieved with the "NORMAL" modes.

A&B TRIGGER MODE: Trigger on 2 unrelated waveforms simultaneously.

ACCURATE METERS: The inherent accuracy of an 8 bit sampled signal is rather poor, specially when only a few "steps" are involved with low signal levels. Likewise, the accuracy of time/frequency measurements can be poor when only a few samples are included in the measurement. This revision attempts to improve on this, as much as is possible with accurate calibration, summing ALL frame transfers (rather than just one frame prior to displaying the result), and where possible using quantization error correction.

LARGE METERS: In addition to the regular types. Fully customizable using same method as the regular meters.

MULTIPLE CONFIGURATIONS: Different program configurations can be saved and be available for recall. Can also be used to save custom meter settings.

MENU SHORTCUTS: One button push shortcuts to access V/div, T/div, Trig source and Trig level

ON SCREEN CHART For quick reference of button functions.

IMPROVED CALIBRATION: Added functions can help to improve calibration. Master clock compensation (typical units are 60-80 off), the use of 3 digits of precision for voltage calibration, an indicator to precisely set the calibration on the center of a step and compensation for battery/power supply variations. While any of these can be skipped as some tend to be tedious to use, they are available for those who want improved results, and it only needs to be done once.

IMPROVED X/Y MODE: Ability to adjust buffer size, as well as a persistence mode help "fine tuning" XY mode to properly display moving or complex patterns.

IMPROVED WAVEFORM GENERATOR: Higher quality waveforms by using up to 10 times greater sampling rates. Also ability to continuously adjust frequency within each range.

ADVANCED GENERATOR FUNCTIONS: Sweep, burst (adjustable from 1 to 1000 sweeps/bursts per sec), white noise, arbitrary, serial code and precision pulse waveforms. Detector mode to display frequency response. Auto synchronization in detector mode between timebases and frequency sweep to display sweep generator frequency response.

BMP LOAD: View saved screenshots.

MAP MODE: displays 200 frames, one per line while modulating color in response to signal amplitude. Useful for example to find missing pulses over an extended amount of time.

PERSISTENCE MODE: allows comprehensive view of a changing signal. Can be set to continually persist, with manual reset, or to auto reset after an adjustable number of frames.

FFT/SPECTROGRAPH FUNCTIONS: Increased gain by up to 48 decibels (in log mode) and increased dynamic range by using 32 bit buffers. Auto, log and manual amplitude scaling. Summing function provides flat frequency response. FFT magnification mode expands frequency range by up to 10000x with 7digit meters and 1PPM resolution. Persistence mode provides peak hold. Envelope display available with spectrograph.

Display improvements

• DISTORTION Visible under certain conditions as increased noise on CHANNEL B caused by error in FPGA programming on pre-HW 2.81 devices fixed.

• SWITCHABLE WAVEFORM AMPLITUDE COMPENSATION allows compensation "steps" to be switched off to improve waveform fidelity.

• WAVEFORM "THICKNESS" ADJUSTMENT

• EXCESSIVE JITTER on the fastest interpolated time bases has been eliminated.

• NON LINEARITY at bottom of screen minimized by shifting ADC operation point.

• AVERAGING and 8x OVERSAMPLING BUFFER MODES can greatly improve display under certain conditions, eliminating jitter for most timebases.

• FULL SPEED OVERSAMPLING (Available with HW2.81 with included W1.1 FPGA update): 72Ms/sec sampling for ALL TIMEBASES.

Additional operating functions

• AUTO-SET adjusts Vranges, Ypositions, timebase, trigger source and meter page automatically.

• CURSOR METER displays position value of V, T or trigger cursors.

• CURSOR SELECT METERS allows measurements to be restricted to an area of the waveform defined with cursors.

• MIN/MAX HOLD functions store peak measured values of a waveform.

• ALARMS can be set to go off if a value exceeds or goes below a cursor position.

• STANDBY control: Engage standby mode manually or shut off standby timer.

• TRIGGER HOLDOFF for the slower time bases.

• SERIAL, I2C and SPI DECODING, showing individual byte/word positions along the waveform as well as a chart of both HEX and ASCII values

• CHART MODE (extended time base up to 10 min/div with averaging or oversampling)

• WAVEFORM INVERT mode for analog channels.

• AUTO SAVE INCREMENTING BUF AND CSV FILES in chart mode allows recording large amount of data.

• SAVE BINARY IMAGE OF ENTIRE ROM for device restoration if ever needed.

• DIRECTORY LIST AND FILENAME EDIT.

SECTION 03

Controls and on-screen menus

Every button, both toggle switches, context actions, menu groups and the status area.

Press timing

BUTTON PRESSES: SHORT PRESS = Press button and do not hold for more than 1/2 second LONG PRESS = Press button and HOLD for more than 1/2 second but NO MORE than 1.5 seconds (buzzer will beep and display flash after 1/2S) HOLD = Press and HOLD button until function appears, after about 1.5 seconds (~3 seconds for calibration menu)

Detailed button operations

1 · Button 1

DOUBLE TAP: ENGAGES AUTO-SET. Press button 1 again to cancel.

SHORT PRESS: TOGGLES HOLD/RUN RESETS MIN/MAX meters in HOLD MODE NOTE: that CHANGING SETTINGS SUCH AS BUFFER MODES, V/div AND TIMEBASES while in HOLD MODE can result in UNINTENDED RESULTS such as incorrect meter readings since updating is suspended. XPOS and CURSOR RESTRICT METER MODE can be used in regular buffer modes while on hold to scroll through, view and measure buffer contents.

LONG PRESS: SHORTCUT TO ANALOG CHANNELS V/DIV MENU ITEM If ALREADY on an analog V/DIV item, will TOGGLE THIS CHANNEL ON/OFF If NO ANALOG channels are on, will TURN ON channel (in FFT mode will only GO to V/DIV. Toggle trace on/off with second press) If CH-A or CH-B menu is active, will go to V/DIV item of that channel if not already on it Otherwise will attempt to go to last channel turned on. If wrong channel is selected when both are on, shift over with right toggle

HOLD BUTTON: DISPLAYS BUTTON FUNCTION REFERENCE CHART Press BUTTON 1 to exit without doing anything Press ANY OTHER BUTTON to exit chart and execute it's normal function

2 · Button 2

SHORT PRESS: Selects AUTO TRIG LEVEL Will default to 1/2 of waveform level from peak to peak. Press again to select 3/4 and 1/4 of waveform level In GEN TRIG MODE WITH SWEEP OR BURST OUTPUT only 1/4 or 3/4 will be available.

LONG PRESS: Selects MANUAL TRIG LEVEL MODE Menu will shift over to TRIGGERING MENU and TRIGGERING LEVEL SUB MENU

HOLD BUTTON: Enters CALIBRATION ROUTINE or ADJUSTS certain MENU DEPENDENT VALUES: With MENU on CH A OR CH B: Enters CALIBRATION MENU With MENU on TIME/DIV SUB MENU WHILE IN FULL SPEED OS MODE (HW2.81-W1.1 FPGA ONLY): adjusts FULL SPEED BUFFER OVERSAMPLING SPEED (Change with left toggle)

With MENU on ANY OTHER than FILE: ADJUSTS ADC WINDOW POSITION

3 · Button 3

SHORT PRESS: METERS OFF > REGULAR METERS > LARGE METERS

LONG PRESS: ENGAGES STANDBY MODE (If generator is not OFF, disables scope providing better waveform quality)

HOLD BUTTON: Selects SAVE CONFIGURATION MENU SHORT PRESS BUTTON 3 AGAIN TO SAVE (Menu will return to prior selection) Default displayed after boot will be CONFIG #0 (boot up config) Change config number (0 to 9) if desired before pressing to save (Active config # in use will be displayed in status area) OR if in filename edit mode, will display last loaded/saved config name for editing.

Press again to save. 4) BUTTON 4 SHORT PRESS: Selects ITEMS (sub-menus) within menus IN METER ITEM SELECTION MODE (Long press right toggle center button to access): SELECTS METER CHANNEL SOURCE

LONG PRESS: SHORTCUT TO TIME BASE TIME/DIV

HOLD BUTTON: Toggles CURSOR RESTRICTED METER FUNCTION "X" appears in status area when function is ON

5 · Left Toggle

: <LEFT-RIGHT> ADJUSTS SELECTED (FLASHING) MENU ITEMS IN METER ITEM SELECTION MODE (Long press right toggle center button to access): CHANGES SELECTED (flashing) METER FUNCTION

LEFT TOGGLE CENTER PRESS: SHORT PRESS: SETS TRIGGERING SOURCE CHANNEL If only 2 channels are ON, TOGGLES BETWEEN THE TWO. If Menu is ON a channel, sets triggering to that channel. If more than 2 channels are ON, acts as a SHORTCUT TO TRIGGERING SOURCE MENU (does not change source) If CH-A AND CH-B ONLY are ON, will cycle CH-A > CH-B > CH-A&B

ENGAGES SPECIAL CONTROLS IN SELECTED FUNCTIONS (Adjustment items will show in notification area): IN WAVE OUTPUT MENU (FREQ RANGE SUB MENU) Toggles WAVE OUT SPECIAL FUNCTIONS (Sweep/burst adj) IN WAVE OUTPUT MENU (% DUTY PWM SUB MENU) Toggles PULSE MODE/PULSE WIDTH ADJ IN CH 1 MENU (COUPLING SUB MENU) : In serial decode (RS232/TTL): Toggles SERIAL DECODE DATA/PARITY ADJUST In SPI decode : Adjusts SPI parameters (shift between parameters with RIGHT toggle CENTER button IN CH 2 MENU (COUPLING SUB MENU): In SPI decode, toggles decode data display chart IN XY MODE: Toggles PERSISTENCE ON/OFF IN T1-T2 MENUS while in SERIAL DECODE: Toggles AUTO (T2-T1) or MANUAL (T2+T1) MODE IN NORMAL (TIME BASE SELECTED) PERSISTENCE MODE , WHILE ON PERST MENU (flashing): TOGGLES RAW (non interpolated) MODE ON > OFF IN FILE MENU: LOADS/SAVES FILE (When loading config, number in use will be displayed in status area, or name at top left of screen) IN CHART MODE (with TIME BASE MENU ON TIME/DIV) Engages BUF or CSV auto incrementing file save select > AVERAGING or OVERSAMPLING buffer mode select > Back to time base menu (change settings with left toggle < >) WITH MENU ON DISPLAY BRIGHT ADJUST: Engages waveform display fast rise adjust (change with left toggle) WITH TIME BASE MENU MODE and AUTO MODE selected: Accesses untriggered auto mode selection (change with left toggle)

LONG PRESS: WITH METERS ON > CHANGES METER PAGES (Note CH-A&B page is NOT CUSTOMIZABLE) CH-A&B > CH-A > CH-B > CH-C > CH-D (Note when changing pages custom selected meter items will revert to defaults. If saving custom meters is desired, save configuration, custom meter settings will be stored along with other parameters)

WITH METERS OFF > Toggles WAVE CALIBRATION ON/OFF ("C" or "U" is displayed in status area)

HOLD BUTTON: TOGGLES TRIGGER HOLDOFF ("T" will show in notification area, position indicating relative delay. Color will be of channel color if in supported mode/timebase, greyed out if not)

6 · Right Toggle

: <LEFT-RIGHT> SELECTS MENUS (HOLD LEFT to go to CH-A, HOLD RIGHT to go to FILE MENU) IN METER ITEM SELECTION MODE: SELECTS METER FOR CUSTOM SETTINGS

RIGHT TOGGLE CENTER PRESS: SHORT PRESS: SELECTS BUFFER SIZE/MODE: SHORT (1 screen wide) > SHORT AVERAGING > 8x OVERSAMPLING > FULL SPEED OVERSAMPLING > LONG (Approx 10 screens wide) IN SPECTROGRAPH MODE SELECTS (IF CH4 MENU BLINKING): SPEC > SPEC + ENVELOPE > ENVELOPE IN WAVE OUT SPECIAL FUNCTIONS MODE (IN NOTIFICATION AREA): CHANGES SPECIAL FUNCTION SELECTION IN SERIAL DECODE AND SPI MODES (IN NOTIFICATION AREA): SHIFT ACCESS TO NEXT PARAMETER WITH MENU ON DISPLAY BRIGHT ADJUST: ADJUSTS WAVEFORM HORIZONTAL THICKNESS WITH MENU IN FILENAME EDIT: RESETS SELECTED CHARACTER TO SPACE WITH MENU ON FFT/MAGNIFY (IN NOTIFICATION AREA): TOGGLES BETWEEN AUTO AND T1 MODE

LONG PRESS: WITH METERS ON : TOGGLES NORMAL MENU SELECTION > METER ITEM SELECTION (For custom meter settings) WITH METERS OFF: TOGGLES STANDBY TIMER ON > OFF WITH MENU IN FILENAME EDIT: RESETS ALL CHARACTERS TO SPACES WITH FILE LIST DISPLAYING FOR LOADING: DELETE SELECTED FILE (press again to confirm)

HOLD BUTTON: TOGGLES MIN/MAX HOLD FUNCTION ON > OFF (Meter MIN/MAX titles will show in WHITE when hold is on, values will show in selected channel color, "H" will also show in status area)

Menu and status reference

STATUS AREA: BELOW CH C (4) IN WHITE U or C - Wave calibration on (C) or uncalibrated (U) H - Hold mode ON for Vmax and Vmin X - Cursor selected meters ON 0-9 - Active config file in present use (0 is boot up config)

1) RUN/HOLD (display only) Displays RUN/HOLD status

<ADJUST ITEMS WITH LEFT TOGGLE> <CHANGE MENUS WITH RIGHT TOGGLE> <CHANGE SUB MENUS WITH SHORT PRESS BUTTON 4>

>>NOTE that once a sub menu ITEM has been selected, it will STAY when leaving and coming back to the menu. This is handy as it allows a quick selection between the items in different menus, however it may be confusing to first time users as the focus may seem to be flinging around randomly from place to place on the screen.

SUB MENUS ITEMS

2-3) ANALOG CH A AND B CH source: OFF > CH(A) > CH A(inv) > CH(A)x10 > CH Ax10(inv) (Displays CH-A=X or x10A=X, CH-B=Y or x10B=Y in XY mode)

COUPLING AC > DC > DT > TL > RS > I2 > SP (AC > DC > Detector mode > TTL serial decode > RS-232 serial decode > I2C decode > SPI decode) NOTE: TL and RS only available with CH-A VERT RANGE 50MV / DIV TO 10V/DIV in 1-2-5 steps

Y POSITION (Bottom left of screen)

4) DIGITAL CH 3 CH source OFF > CH(C) (NOTE THAT CH 3 IS DISABLED if trig source is A&B)

Y POSITION (Bottom left of screen)

5) DIGITAL CH 4 CH source OFF > CH(D) > (A+B) > (A-B) > (C&D) > (C or D) > DAT_A > DAT_B > DAT_C > DAT_D > MAP > SPEC > hFFT_B (Hann window) > sFFT_B (Summing) > hFFT_A (Hann window) > s (Summing) FFT CH-A (NOTE THAT CH 4 IS DISABLED if trig source is A&B)

FFT GAIN (In notification area, only in FFT/SPEC Modes) LOG>AUTO > 0db > +6db >.....> +36db > +42db

Y POSITION (In all other Modes, selection blinks at bottom left of screen)

6) GENERATOR OUTPUT WAVE TYPE OFF > ARBTR > UART > SINE > TRIANGLE > SAWTOOTH > SQUARE > PWM > NOISE

FREQUENCY (RANGE) PWM= 1HZ TO 8MHZ NOISE= 1HZ to 1MHZ ALL OTHERS= 1HZ TO 200KHZ

LEVEL/DUT% PWM= DUTY 1% TO 99% ALL OTHERS= LEVEL 0 TO 2.6V P-P

7) TIME BASE SCAN MODE TRIGGERING OFF > AUTO > NORMAL > NORMAL W/CLEAR > PERSISTENCE > SINGLE > XY (NOTE that only NORM will show with A&B source) TIME/DIVISION 1SEC TO .1uS In 1-2-5 steps (10 MINS to 100 mS / DIV in CHART MODE)

X POSITION (Selection blinks at right bottom of screen) Shifts hor position of triggered waveform (In LARGE BUFFER MODE, notification area shows window POSITION within buffer) WHEN PAUSED IN AVERAGING OR OVERSAMPLING BUFFER MODE only moves TRIG CURSOR

PERS FRAMES (In notification area, only in PERSISTENCE MODE) PERSISTENCE FRAMES: CONTINUOUS > VARIABLE FROM 1 TO 256 FRAMES

XY SAMPLES (In notification area, only in XY MODE) BUFFER SIZE FOR XY MODE= VARIABLE FROM 256 TO 4096 SAMPLES

8) TRIGGERING MODE TRIG SOURCE CH-A > CH B > CH C > CH D > CH-A&B (Selecting CH-A&B as SOURCE disables digital channels)

TRIG TYPE + TRANSITIONS > - TRANSITIONS > LEVEL GREATER THAN CURSOR > LEVEL LESS THAN CURSOR > LOW LESS THAN DELTA T1-T2 > LOW GREATER THAN DELTA T1-T2 > HIGH LESS THAN DELTA T1-T2 > HIGH GREATER THAN DELTA T1-T2 > GEN

TRIG LEVEL Adjusts triggering cursor level (NOT AVAILABLE IN AUTO TRIG MODE)

TRIG DELAY (In notification area, if delay is enabled, only in NORMAL MODES, 1mS and slower and in single frame buffer mode only) Adjust trigger delay in steps of 1 major div. Shifts "T" indicator.

9) VOLTAGE CURSORS V1 Adjust top VOLTAGE CURSOR (Can be used to adjust parts of waveform to be included in meters in CURSOR SELECT METER MODE) V2 Adjust lower VOLTAGE CURSOR (Can be used to adjust parts of waveform to be included in meters in CURSOR SELECT METER MODE) In FFT mode, shows reference db LEVEL.

V SOURCE Selects CH-A or CH-B as SOURCE to display in DELTA V display at top right of screen.

10) TIME CURSORS T1 Adjust left TIME CURSOR position (Can be used to select parts of waveform to be displayed in meters in CURSOR SELECT METER MODE, also adjusts START POSITION in SERIAL DECODE MANUAL MODE) In FFT mode, shows FREQUENCY at position of cursor. T2 Adjust right TIME CURSOR position (Can be used to select parts of waveform to be displayed in meters in CURSOR SELECT METER MODE, also adjusts BAUD RATE in SERIAL DECODE MODE)

11) NOTIFICATION AREA/FILE MENU LOAD/SAVE LOAD FILE > SAVE FILE FILE # CHANGE FILE NUMBER (same number unit used for all types) FILE TYPE CONFIGURATION FILE > ARB > UAR > ROM (BIN) > BMP > DAT > BUF > CSV (CSV and ROM cannot be loaded), ARB and UAR only in LOAD MODE) LIST/EDIT If LOAD FILE IS SELECTED, FILE LIST IS DISPLAYED. IF SAVE FILE IS SELECTED, FILENAME EDIT FUNCTION IS DISPLAYED.

SECTION 04

Acquisition and triggering

Set channels, acquisition buffers, timebase, trigger logic, persistence and chart capture.

AUTO-SET: Double tap BUTTON 1 QUICKLY (< 1/2 sec). Program will turn on any channel with a signal, set V ranges, Y positions, timebase, trigger source and meter page for a usable display. If no signal is present, autoset will pause and wait for a signal, notification will show status.

If only DC is present, function will display it, look for a trigger for a few seconds, then terminate with a default timebase of 5mS/div. Press button 1 again at any time while function is active to cancel and revert to prior configuration.

Autoset will use the following settings upon completion: -AC or DC coupling, any other selection will default to DC -Trig type >Vt. -Menu will be preset to Time/Div.

Other settings will persist when using autoset and can be used to modify results, for example: -A&B trig mode will persist if both channels have signals. -Buffer mode and any triggered timebase mode active before autoset will be restored upon completion.

Analog and digital channels

VOLTS/DIV SHORTCUT: LONG PRESS BUTTON 1 to access either CH A or CH B VOLTS/DIV sub menus. Also can be used to TOGGLE CHANNEL ON/OFF. See BUTTON FUNCTIONS section for more info. Channels A and B are analog channels and have coupling and Y (vert) sensitivity range sub menus. Change SUB MENUS with BUTTON 4. ALL have Y position sub menus.

CHANNEL C is a simple digital channel. CHANNEL D has MULTIPLE FUNCTIONS.

In addition to selecting the channel D input, it can display the sum and difference of CH A and CH B, AND and OR LOGIC between CH C and CH D, traces saved in DAT file formats (after loading the DAT file, select REC_A, B, C or D), and selects FFT, SPECTROGRAPH and MAP displays.

NOTE that SPECTROGRAPH and MAP displays WILL NOT SHOW ANYTHING unless either CH A or CH B are ON (CH A has precedence if both are on)

Waveform generator

ADJUST WAVE TYPE by shifting output menu. OFF > ARBT > UART > SINE > TRIANGLE > SAWTOOTH > SQUARE > PWM > NOISE. Sine, triangle, sawtooth and square range is from 1hz to 200khz and can be ADJUSTED FOR AMPLITUDE by shifting sub menu to Vo (volts output).

Best quality will be at maximum amplitude as internal digital background noise will become noticeable at the lowest settings. PWM (pulse width modulation) can be adjusted for DUTY% (Vo menu changes over to duty%) but cannot be adjusted for amplitude. Frequency for this wave type ranges from 1hz to 8Mhz.

ARBT (arbitrary) wave range depends on number of samples, see "USING ADVANCED GENERATOR FUNCTIONS". SHIFT FREQUENCIES (RANGES with variable control) with FREQUENCY SUB MENU. Shift to OFF when not using generator, some of the advanced functions take a considerable amount of processor resources and can slow down main scope operation.

VARIABLE: CONTINUOUSLY SHIFT FREQUENCIES within each range by PRESSING LEFT TOGGLE CENTER BUTTON while FREQ RANGE MENU IS FLASHING.

Frequency is displayed in notification area. (Shift back by pressing again) Adjust with LEFT TOGGLE STEPS of approx +/- 0.7%, increasing at the higher frequency ranges due to the limitations of the scaling dividers in the system.

RANGE of approx +/- 1 octave above and below the selected range (4:1 ratio overall), the higher frequency ranges having lower high frequency limits due to sampling rate limitations of the internal DAC. Selected frequencies will be RESET to the range default when changing ranges.

NOTE THAT FREQ ADJ CAN BE CHANGED TO SWEEP OR BURST ADJ (see "advanced gen functions").

SAMPLES PER PERIOD is 720 samples for frequencies of 2.5Khz and below, 360 for 5khz range, 180 for 10khz and 72 for the 20khz range, due to limitations of the internal DAC. Frequencies of up to 200Khz can be selected with reduced quality but can still be useful. NOTE that a DC OFFSET is present at the wave out connector, regardless of the wave type or level. This may be desirable in some cases such as driving logic circuits but in other cases may need to be filtered out externally.

Timebase and acquisition modes

TIME/DIV SHORTCUT: LONG PRESS BUTTON 4 at any time to access TIME/DIV sub menu. NOTE that in A&B TRIGGER MODE, mode will be LOCKED in NORML. Previous selection will be restored upon exiting A&B source.

<TrOFF> Triggering and synchronization completely disabled, device will start new scan as soon as it reaches the right side of the screen. NOTE that in TrOFF mode ONLY, extended time base is available (1 sec/div to 10 min/div, see CHART MODE).

<AUTO> Scans in normal triggered way. If triggering is lost, after an appropriate delay, will auto scan. Note that at low scan speeds, If monitoring relatively low frequencies, triggering may be lost as it may enter auto scan before being triggered again. If this happens, switch to <NORMAL> mode.

WHILE IN AUTO trig mode, with MENU SELECTION ON AUTO (AUTO blinking): Press LEFT TOGGLE CENTER BUTTON to toggle UNTRIGGERED AUTO MODE SELECTION Change with LEFT TOGGLE. Setting is saved with config save. (Only available with HW2.81 and FPGA W1-W1.1)

Untriggered AUTO choices

AUTO trig CONT WAVE= Pauses each frame before displaying while not triggered so waveforms can be represented correctly. Data is not saved while paused for displaying so at first trigger, pre-trigger data is lost.

AUTO trig FAST RELEASE= Lets data continuously overwrite even if display can't keep up so at first trigger event while untriggered, pre-trigger data can be displayed. However, continuous overwriting can result in broken waveform display while not triggered at faster timebases.

NOTE that with either setting, some trigger events that occur while in untriggered display may be properly captured, so these may be missed. Use NORMAL modes to properly capture slow or occasional events if necessary.

NORMAL MODES: <NORML> Provides best triggering as after scan is complete, the device just waits for another trigger. <NorCl> NORMAL WITH CLEAR: <NORML> mode leaves last scanned frame visible on screen while waiting for another trigger. <NorCL> clears last displayed frame after a short delay. This can be useful for example to monitor a series of identical pulses. In normal mode, new pulses would just overwrite the old one and it may not be apparent a new event has occurred.

Persistence mode

SELECT <PERS> IN TIME BASE MODE: All traces will stay displayed on screen until manually reset if in CONTINUOUS MODE. SELECT CONTINUOUS OR AUTO RESET NUMBER OF FRAMES: With menu in TIME BASE, cycle items with BUTTON 4 to how FRAMES= in notification area. Change with LEFT TOGGLE. At extreme left will be CONT (continuous).

Shift to right to select: AUTO and NUMBER OF FRAMES to show until auto resetting. RESET (if in continuous mode) by engaging ANY button or toggle. Freeze display with HOLD, then press again to reset. PERSISTENCE MODE acts as a NORMAL mode, so will not show any display (or change) until triggered.

TOGGLE RAW MODE by SHORT pressing LEFT TOGGLE CENTER BUTTON. Raw mode turns off all interpolation, so that only data POINTS are displayed, with no lines connecting them.

This can be useful in improving resolution when many waveforms are close to each other, since interpolation lines have a tendency to quickly cover everything up if too close. The first frame in RAW mode will nevertheless be displayed with interpolation to improve graphics, only subsequent frames will be displayed as data points.

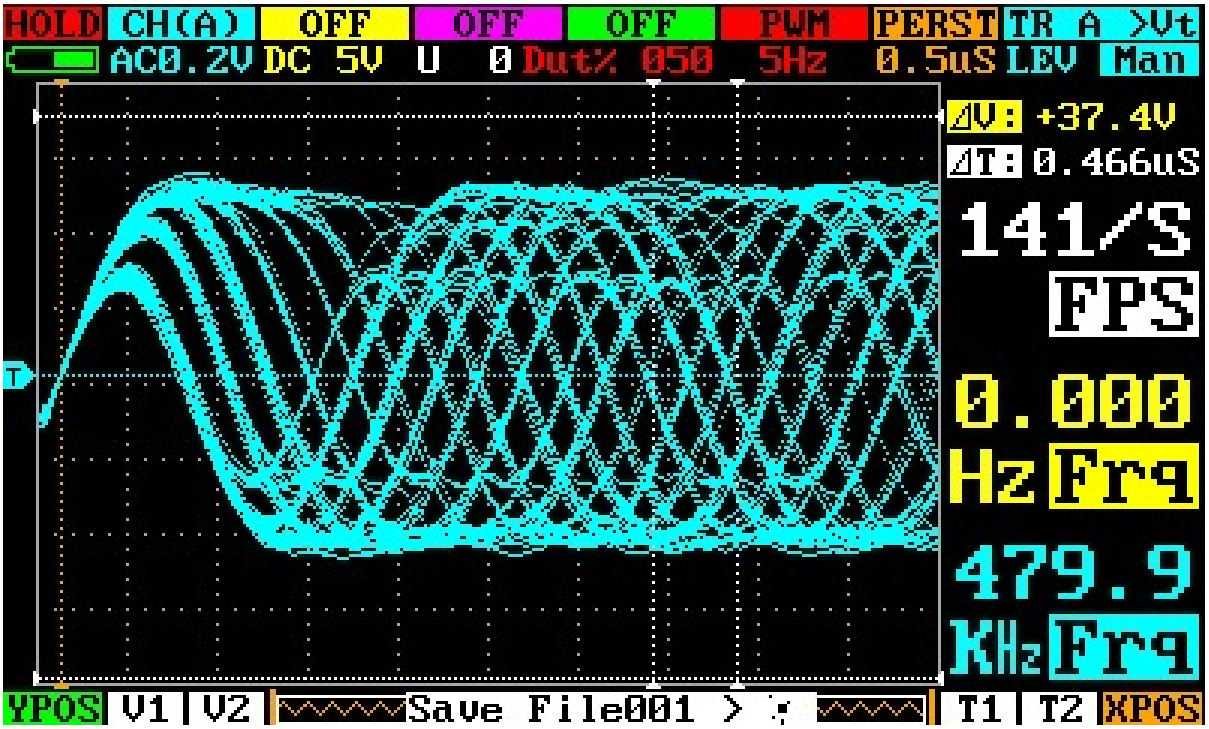


FIG. 03 Persistence mode: RF eye pattern in RAW data mode.

SINGLE SELECT <SINGL> Display pauses after capturing a single frame. Reset with SHORT PRESS BUTTON 1 NOTE that single mode will default to large buffer mode except in averaging and oversampling buffer mode where it will capture only 1 frozen frame (XPOS will shift trig point, not window)

X/Y display

SELECT <XY> IN TIME BASE MODE: Channel A is displayed along X axis, and channel B along the Y axis. While in TIME BASE MENU, change items (BUTTON 4) to show SAMPLES= in notification area. Change number of samples with LEFT TOGGLE. Range includes from 256 to 4096 samples, which corresponds to a frame rate of around 50 to 6 or so FPS.

Faster frame rates (fewer samples) will result in a more "fluid" display with moving patterns, the faster a pattern is moving the faster is the frame rate necessary to follow it correctly. However, more samples improves the graphics, and if there are not enough samples the pattern may not be complete, showing a gap.

If there are too many samples, the pattern may fold upon itself, showing multiple traces. In such a case shifting to a faster time base will correct this. A bit of experimentation with TIME BASE SETTINGS and SAMPLE NUMBERS will quickly show optimum settings for each display. For STATIONARY PATTERNS, PERSISTENCE MODE can be used.

SHORT press LEFT TOGGLE CENTER BUTTON to toggle mode. This will improve graphics as dots will quickly populate the display to form lines. To SHIFT PATTERN POSITION use CHA XPOS to shift HORIZONTALLY and CHB XPOS to shift VERTICALLY

Buffer modes, averaging and oversampling

Short press RIGHT TOGGLE CENTER BUTTON to cycle between 4 buffer modes: Color of zigzag in notification area will change with mode Regular single window buffer (orange) > Averaging single window buffer (green) > 8x Oversampling single window buffer (red) > Full speed oversampling buffer (blue)> Large buffer mode (orange) Rectangle will be orange in OS modes, white in non-OS modes.

In any of the single frame buffer modes, notification area rectangle will occupy the whole section, while in large buffer mode an appropriate sized rectangle will show the present position of the visible frame within the buffer. XPOS (sub menu of the time base menu) will shift the visible frame position within the buffer.

REGULAR SINGLE WINDOW BUFFER is used to get the fastest frame refresh rates. Scan ends at the end of the screen and the program waits for another trigger. Single window modes are specially useful at the slower time bases, where filling the entire buffer would take a considerable amount of time, slowing refresh rates to an unacceptable level.

AVERAGING SINGLE WINDOW BUFFER ends scan after screen is displayed like the regular mode but accumulates multiple additional samples and averages the result before displaying, effectively producing a low pass filter with a cut off point just above each timebase's bandwidth.

Produces very clean, jitter free waveforms with minimal noise at the expense of a somewhat slower frame refresh rate. Using this buffer mode (as well as oversampling buffer mode, since FFT defaults to averaging in that case) with FFT functions will help reduce aliasing when out of band high frequency components are present.

OVERSAMPLING SINGLE WINDOW BUFFER accumulates multiple samples before displaying and records positive and negative peaks in the waveform before displaying them as a vertical bar, effectively extending the frequency response to above the timebase's bandwidth. Reduces aliasing artifacts and eliminates jitter in the displayed waveform.

Use this to display high frequency waves or HF wave components well above the sampling rate, when low frequency components need to be observed, such as monitoring modulation on an RF carrier. NOTE that in this mode FFT FUNCTIONS use AVERAGING, as in the averaging buffer mode since oversampling is not suitable.

XPOS in AVERAGING AND OVERSAMPLING MODE works a bit differently than in regular buffer modes. Instead of shifting window position and/or adjusting the size of the buffer, buffer size is fixed, and XPOS in these two modes shifts the triggering position. When paused, changing XPOS will only change the trigger vernier and not shift the window.

Full-speed oversampling

(Only available with HARDWARE V2.81 when used with included FPGA V1.1 update) A single window buffer mode which sets the sampling speed at a default of 72M samples/second for ALL TIMEBASES, except for very lowest of 50Ms/div and lower for which the speed is gradually reduced to maintain a 60,000 x oversampling ratio with the timebase.

Actual speed can be adjusted gradually in steps down to 2Ms/sec by holding down BUTTON 2 for MORE THAN 3 SECONDS while menu is on TIME/DIV. Save with config 0. Reducing speed can be useful in some cases for example to reduce noise.

How this mode works: Incoming waveforms are sampled at a continuous 72Mhz rate, regardless of timebase.

Different timebases are then generated by dividing the samples, and storing the most positive and most negative point within each of these divided periods, and representing these as one single sample on screen, which results in a vertical bar between these minimum and maximum points, the only way complete waveforms can be represented within one sample.

This provides a display much like an analog scope, and extends the bandwidth to the full capability of the hardware without aliasing, regardless of timebase. Will give much better display of RF modulation envelopes and other high frequency events when viewed at slow timebases, without aliasing.

Unlike the averaged and regular (8x) OS mode, which sample triggering at 8x and store the oversampled synchronized waveform in memory, data is oversampled on the fly in this mode, as there is not enough memory to store the waveforms at these sampling speeds, so no memory exists of the triggered and synchronized OS waveforms other than the max and min values.

Therefore, triggering only occurs at each display sample point as in regular modes so this mode does not benefit from the jitter stabilizing effect of the stored OS modes.

METERS and FFT FUNCTIONS get an averaged or peak "screen derived" value in this mode as oversampling is processed in the FPGA and is happening at too fast a rate for the program to access and process, so these may not be as accurate as with other buffer modes, particularly when displaying a large amount of waves (or when an FFT display peak is towards the right of the screen).

NOTE: OVERSAMPLING and AVERAGING MODES from 100x to 600x are available for all hardware versions for CHART MODE (see chart mode for more info)

Large buffer mode

records all 4096 samples and stores them in a buffer. Use XPOS sub menu of TIME BASE MODE to shift the window view along the buffer. Since all 4096 samples need to be obtained before the display refreshes, this can take a considerable amount of time, specially in the slower timebases. Gives a record of approx 10 windows width. METER ACCURACY, particularly time based meters, will benefit from using this mode, since many more samples are available for calculations.

NOTE that in SPECTROGRAPH MODE, GEN OUTPUT FREQ, BACKLIGHT BRIGHTNESS ADJUST AND SERIAL DECODE MODES where items need to be selected in the notification area, right toggle center press has a different function.

Trigger source

TRIGGERING SOURCE SHORTCUT: SHORT PRESS LEFT TOGGLE CENTER BUTTON when NOTIFICATIONS ARE NOT DISPLAYED, to set triggering source. An attempt will be made to set source to a relevant channel. See BUTTON FUNCTIONS for more info. NOTE that this button has several ALTERNATE FUNCTIONS depending on the menu selected. In TRIGG MENU select TR A, B, C, D or A&B for triggering source channel. Selecting A&B will lock time base mode in NORML. Previously selected time base mode will return upon exiting A&B mode.

Trigger level

MANUAL TRIGGERING LEVEL SHORTCUT: LONG PRESS BUTTON 2 to put trigger level mode in MANUAL. Menu will shift to LEVEL and will change to MANUAL. Adjust trigger point with LEFT TOGGLE. As long as mode is in MANUAL, item can be selected to manually set trigger level.

AUTO TRIGGER LEVEL: SHORT PRESS BUTTON 2. MAN in trig sub menu will change to 1/2 (sets level to the middle of the waveform) SHORT PRESS AGAIN to set to 3/4, then 1/4 of wave level.

Trigger types

Change TRIGG MENU TYPE ITEM: LEVEL BASED: >Vt <Vt Triggers when wave crosses over the TRIGGER CURSOR to above or below. EDGE BASED: POS EDGE, NEG EDGE Triggers when wave crosses over the TRIGGER CURSOR on positive or negative edges TIME BASED: LOW LESS THAN DELTA T1-T2: When the period of the waveform as it crosses the trigger cursor DOWN to the time it crosses the cursor BACK UP is SHORTER than the DIFFERENCE BETWEEN THE T1 AND T2 CURSORS ON SCREEN, the device will trigger when COMING BACK UP.

LOW GREATER THAN DELTA T1-T2: When the period of the waveform as it crosses the trigger cursor DOWN to the time it crosses the cursor BACK UP is LONGER than the DIFFERENCE BETWEEN THE T1 AND T2 CURSORS ON SCREEN, the device will trigger when COMING BACK UP.

HIGH LESS THAN DELTA T1-T2: When the period of the waveform as it crosses the trigger cursor UP to the time it crosses the cursor BACK DOWN is SHORTER than the DIFFERENCE BETWEEN THE T1 AND T2 CURSORS ON SCREEN, the device will trigger when GOING BACK DOWN.

HIGH GREATER THAN DELTA T1-T2: When the period of the waveform as it crosses the trigger cursor UP to the time it crosses the cursor BACK DOWN is LONGER than the DIFFERENCE BETWEEN THE T1 AND T2 CURSORS ON SCREEN, the device will trigger when GOING BACK DOWN.

GEN: Triggers after space between sweeps or bursts from generator, on first pos transition if above wave center or neg transition if below. If generator is set to a continuous wave type, triggering in this mode is set to > VT.

Note that it makes NO DIFFERENCE WHERE THE T1 AND T2 CURSORS ARE ON SCREEN, only the SPACING BETWEEN THEM. They can be positioned wherever it can be useful to gauge waveform width. Also note that if the time base is changed, the cursors will remain in the same relative spacing, while the "LOW/HIGH TIME" will change, potentially disabling triggering.

Dual-channel A+B trigger

Selecting A&B as a trigger source will allow triggering on two completely different (frequency unrelated) signals simultaneously. ONLY NORMAL TRIGGERING MODE is compatible with A&B source: Selecting A&B source will LOCK the device in NORMAL mode. DIGITAL CHANNELS also will be DISABLED. Previous settings will be restored when exiting mode.

CHANGING TRIGGERING LEVEL either in AUTO or MANUAL mode will affect both A and B trigger cursors equally. To change each independently shift SOURCE TO EITHER A OR B, change the level for that channel then return to A&B mode. Trig cursors will remain where set relative to each other.

NOTE that selecting A&B trigger source will DISABLE AVERAGING and OVERSAMPLING modes.

Delayed trigger

ONLY AVAILABLE IN NORMAL MODES AT TIME BASES OF 1mS/DIV AND LOWER IN SINGLE FRAME BUFFER MODE HOLD LEFT TOGGLE CENTER BUTTON to toggle mode. "T" shows in notification area on buffer display at approximate position of delay. If in supported mode/time base, indicator will show in color of triggering channel, otherwise will be grayed out.

TO CHANGE DELAY: From TRIGG menu change items to select blinking "T" indicator. Move LEFT TOGGLE right to increase delay or left to decrease in steps of 1 (major) screen division. NOTE that at faster time bases or different triggering modes, much of the functionality of a delayed trigger can be had using the time based trigger types.

At time bases faster than 1mS/div timing integrity is lost, as frame rates cannot keep up with incoming signal, so adding extra delay becomes meaningless. At the very fastest time bases, for example, only a very small portion of the incoming data stream is captured.

The lost data period is known as "Dead Time" and is inherent in all but the most expensive instruments available, with more expensive devices offering faster display frame rates and less dead time.

Chart mode

For extended time base of 100mS/div to 10min/div, SELECT <TrOFF>, then shift time base to greater than 1S. This ENGAGES CHART MODE. Time/div display turns WHITE as an indicator. Mode is NOT TRIGGERED and will CONTINUOUSLY RUN until PAUSED. RESET recording position to extreme left of screen by pressing <PAUSE> then <RUN>.

Yellow cursor at top of screen indicates present recording position. NOTE that when sampling reaches extreme right of screen (or end of buffer in full buffer mode) recording will then start over and overwrite continuously.

In FULL SIZE BUFFER, window will scroll when sampling reaches to the right. Shifting XPOS back to view buffer will stop the window scrolling, shifting it forward to the advancing sampling position will allow it to resume. XPOS can be used to move down the buffer whether in RUN, HOLD or SAVED BUF FILE DISPLAY. Also note TIME BASED METERS ARE DISABLED IN CHART MODE.

The very start of the trace can be immediately observed since this mode is not triggered, unlike triggered modes where the pre-trigger 5 screen divisions have to be captured before new data is displayed.

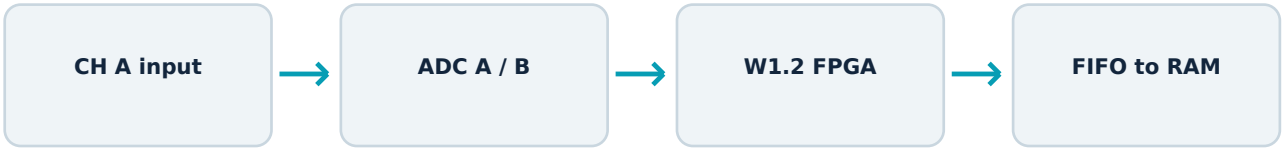
AUTO SAVE INCREMENTING BUF OR CSV FILES: For saving to disk continuously, new incrementing BUF or CSV files can be automatically saved at end of each acquired buffer while in full buffer chart mode. With menu on TIME BASE TIME/DIV press LEFT TOGGLE CENTER BUTTON to access auto file save select. Change with LEFT TOGGLE. Files will start with present # in file menu and will overwrite without notification. IN THE EVENT OF A WRITE FAILURE WHILE AUTO SAVING (eg: drive is full) Auto save will be disabled.

CHART mode is OVERSAMPLED at ratios ranging from 100x at 0.1S/div to 600x at 10Min/div. Access AVERAGING or OVERSAMPLING (peak hold) select menu with LEFT TOGGLE CENTER BUTTON while on TIME BASE TIME/DIV > AUTO FILE SAVE > AVERAGING/OVERSAMPLING select. Change with LEFT TOGGLE. Selection is saved in config files.

144 MS/s acquisition

SECTION 04A / NEW IN WILDCAT 6.6

The 144 MS/s mode is a dedicated high-speed acquisition path for DS203 HW 2.81/2.82 with the FPGA W1.2 image installed. Two physical ADCs are interleaved to acquire one analogue input: CH A. It is not a two-channel 144 MS/s mode.



B analogue input is not sampled in 144 mode. The two ADCs alternately sample channel A.

Fast facts

Item	144 MS/s behavior
Analogue signal	CH A only; analogue CH B is unavailable until returning to Normal.
Sample rate	144 million interleaved samples per second, nominal 6.94 ns between samples.
Digital inputs	CH C and CH D remain 72 MHz digital inputs; their timing is aligned separately.
Timebase	100 ns/div, 200 ns/div or 500 ns/div.
Acquisition record	4096 FPGA words are copied to 16 KiB RAM; after 2 prefix words, 8188 analogue samples are valid.
Calibration	Measured ADC B-to-A relative correction plus absolute CH A gain/offset; profiles depend on saved calibration.

Important. This is a recorded-frame mode, not a continuous SCAN/roll display. A separate low-buffer or faster-display mode has been discussed, but it is not part of the firmware described here.

Required hardware

- DS203 HW 2.81 or 2.82.
- FPGA W1.2 correctly programmed; firmware performs an image/identity check before entering.
- A valid Normal/144 calibration is strongly recommended for accurate voltage values.

Entering, exiting and controls

SECTION 04A / 144 MS/s

Enter from the Normal acquisition-mode cycle

- Switch to CH A AC or DC coupling; complete any active file operation or incompatible function.
- Use the right rocker centre press to cycle the Normal buffer / oversampling modes. With FPGA W1.2, the 144 entry appears at the wrap point of this cycle.
- When activated, the instrument changes to the 144-specific acquisition UI. The main screen retains much of the familiar Normal menu layout.

Return to Normal

- Press the right rocker centre while in 144 mode. The firmware aborts an active capture, frees its buffer ownership and restores the Normal acquisition context.
- No reboot should be required. Any 144-only timebase is converted back to the saved Normal context.

Keys in 144 mode

Control	Action
Button 1	RUN/HOLD as in Normal. SINGLE mode holds after its accepted trigger.
Button 2 (short)	Cycle automatic trigger-level selection for analogue CH A.
Button 2 (medium)	Select manual trigger level on the analogue CH A trigger.
Button 2 (long, approx. 3 s)	With CH A/B menu selected, open the calibration wizard.
Button 3 (short)	Cycle meter presentation (full, compact, hidden).
Button 3 (held)	Open SAVE CFG; standby action is intentionally disabled in 144 mode.
Button 4	Move through the active menu fields.
Left rocker	Adjust selected menu value or cursor/trigger item.
Right rocker, side buttons	Move through the available menu groups.
Right rocker, centre	Exit 144 back to Normal.

Context matters. The exact interpretation of a key can change with the selected menu and duration; use the screen labels rather than relying on a shortcut from older 6.5b versions.

Signal, timebase and trigger

SECTION 04A / 144 MS/s

Analogue signal path

Both ADCs are routed to CH A. Sample slots alternate B then A in the recorded stream. The firmware applies a stored B-to-A correction to the B samples before applying the absolute CH A calibration. Do not connect or expect an independent CH B waveform in this mode.

Timebase and positioning

Timebase	Use
100 ns/div	Fastest view for sub-microsecond events.
200 ns/div	Intermediate view with more context.
500 ns/div	Widest supported window in the 144 renderer.

Horizontal positioning operates on the completed, fixed record. The requested XPOS may be clamped to the valid data extent; the status and time cursors use the effective recorded geometry. Vertical range, vertical position, AC/DC coupling and x1/x10 probe scale use familiar CH A controls.

Trigger modes

- AUTO: attempts a hardware trigger, with a bounded wait; if no valid event arrives, a non-triggered acquisition can provide a fresh waveform.
- NORM: waits for a qualifying event; the UI can show WAIT.
- SINGLE: captures one qualified event and then enters HOLD.
- Analogue trigger source is CH A. Digital trigger sources CH C and CH D are supported with their own alignment handling.
- Trigger criteria include analogue edge/level and supported time-based pulse criteria. Effective trigger position depends on FPGA delay and sample phase.

No imaginary trigger. A trigger threshold outside the ADC guard range is not silently clamped to a false hardware level. AUTO may show an untriggered frame instead; NORM/SINGLE require a valid event.

Display, measurements and FFT

SECTION 04A / 144 MS/s

Waveform display

The CPU transfers a complete 4096-word FPGA record, validates it, and then publishes the frame to the display pipeline. The screen redraws changed trace/UI content and retains a valid previous frame while appropriate. Raw samples and their capture-time calibration belong to one acquisition epoch; a new DAC command must not reinterpret old data.

Measurement and cursors

- CH A analogue measurements include DC mean, RMS, peak-to-peak, min/max and supported time/frequency/duty measurements. Saturation warnings appear as ACLIP-/ACLIP+/ACLIP*.
- CH C/D can provide digital time/frequency/duty/period style measurements when valid samples are available.
- T1/T2 and V1/V2 cursors use the actual completed record and correct 144 MS/s timescale. Cursor readout and visible window remain consistent after panning.
- Invalid or pending captures may leave a previously accepted display frozen. Do not treat a frozen image as a new measurement.

FFT

The existing 512-point FFT can be applied to the completed 144 frame. FFT availability depends on source/mode, the timebase, and sufficient valid samples. The 144 UI shows its labels and range controls using the existing fourth-channel menu region.

What is different from Normal

Normal mode	144 mode
A and B analogue channels can both be independent	A only: both ADCs share CH A.
SCAN/roll and several short-buffer options	Completed-frame FIFO capture; no separate 144 SCAN mode.
Broader timebase selection	Three very fast timebases.
Normal ADC calibration	Additional interleaved B-to-A calibration and FPGA image requirement.

Files, generator and limits

SECTION 04A / 144 MS/s

Saving and loading

- CFG: use the normal FILE menu or Button 3 long-press shortcut for SAVE CFG. File 0 updates the boot calibration envelope carefully; other CFG records retain normal configuration semantics.
- CSV: a valid 144 frame can be exported as CSV. The export follows the recorded sample time, ADC A/B slot order and capture-time calibration.
- BUF: the code path does not expose a standard 144 native BUF save operation. Do not assume that a Normal BUF contains an interleaved 144 recording.
- WPT/WPB: calibration is stored as redundant, verified files named by device serial number. See the revised calibration section.

Generator

The built-in waveform generator is still available subject to the same resource ownership rules as Normal. Configuration and UART/file operations may reserve shared buffers. A conflicting resource displays a busy/unavailable message rather than proceeding with a damaged record.

Features deliberately restricted while entering 144

Condition	Status
FPGA not W1.2 / image invalid	144 mode refused; stay in Normal.
Sweep/burst generator operation	Some active sweep states block 144 entry.
Downconversion, Chart, XY, calibration wizard already active	144 entry refused until the conflicting mode ends.
High-speed refresh without complete RAM transfer	Not implemented in this firmware.
Independent analogue CH B at 144 MS/s	Not possible with this analogue routing.

Preserve your data. Complete a pending save/export before switching modes. HOLD keeps a valid completed epoch, but exiting 144 releases its workspace to Normal.

144 quick checks and troubleshooting

SECTION 04A / 144 MS/s

No 144 entry in the right-rocker cycle

- Check HW 2.81/2.82 and FPGA W1.2 programming. W1.1 supports legacy functions but not this validated 144 entry.
- Ensure CH A has AC or DC coupling and disable incompatible processing (Chart/XY/downconversion/sweep).

WAIT or no fresh trace

- Try AUTO mode; its fallback is intentionally a non-triggered frame when no valid threshold crossing occurs.
- Set an in-range trigger level and verify source selection (A, C or D). NORM and SINGLE are not guaranteed to update without an event.

Wrong level, step artefacts or clipping

- Use the new common Normal A+B and interleaved B-to-A ZERO/DC REF calibration.
- Look for the clip indicator and reduce input amplitude or change vertical range. The two interleaved ADCs should not be interpreted as independent analogue channels.

CSV unavailable

- First obtain a completed, valid frame. A still-pending or failed capture cannot be exported as new data.

Refresh seems slow

The 144 sample clock is not the LCD frame rate. The host must wait for an event, transfer the full FIFO record and draw the waveform. The current implementation has no user-selectable small-frame/streaming 144 mode. Do not report this as an existing feature.

Source check. This section is based on the Wildcat V6.6 code at GitHub commit 4df9eba, especially Wildcat144.c, Capture144.c, Frame144.c, UI144.c and FpgaProfile.c.

SECTION 05

Meters, cursors and files

Configure measurements, use cursor-based readings, and save or reload captured data.

Meters

Select meter mode by SHORT PRESSING BUTTON 3. Cycles NO METERS > SMALL METERS > LARGE METERS. In all meters the color displayed is the color of the source channel. Meters displaying in WHITE are either GLOBAL, not relating to any particular channel (display brightness, battery voltage, etc) . HOLD MODE for V MAX AND V MIN meters also show in WHITE. TIME BASED METERS ARE DISABLED IN CHART MODE.

In SMALL METER MODE: The 5 top meters will always be, from the top: -DELTA-V: Shows the difference in voltage, relating to the present selected range, between the V1 and V2 cursors.

Change the SOURCE CHANNEL for this while in the V cursor menu (shift RIGHT TOGGLE until V1 or V2 at the bottom of screen flashes) Then use BUTTON 4 to cycle V1 > V2 > DELTA V. When DELTA V FLASHES, you can use the LEFT TOGGLE to select either CH A OR CH B.

The display will change to the source color for the selected channel. -DELTA-T: Shows the difference in time, relating to the present time base, between the T1 and T2 cursors. -FREQ: (Alternate to DELTA-T, change with left toggle after selecting with button 4) Shows frequency when T cursors span one wavelength. -BACKLIGHT: Shows the display brightness level.

Select this using the RIGHT TOGGLE. When this meter flashes, backlight can be adjusted with the LEFT TOGGLE. -VOLUME: Shows the level of the buzzer. Select this using the RIGHT TOGGLE. When this meter flashes, buzzer volume level can be adjusted with the LEFT TOGGLE. -BATTERY VOLTAGE: For display only,

In LARGE METER MODE: -Only DELTA-V and DELTA-T will persist on all meter pages, except for the preset page showing only 2 large meters which will also show the same small 5 meters as described above.

Meter pages

Change between the 5 meter "pages" by LONG PRESSING LEFT TOGGLE CENTER BUTTON. One page is preset and cannot be changed and will show only 2 meters in large meter mode. LONG PRESS again to access 4 customizable pages, each preset to favor one of the channels.

Customizable meters

BOTTOM 9 REMAINING METERS (top 3 in large meter mode) can have their items and source changed. LONG PRESS RIGHT TOGGLE CENTER BUTTON to change focus from the regular menus to the meters. Use RIGHT TOGGLE to select meter to customize. LEFT TOGGLE will CHANGE METER ITEMS. SHORT PRESS BUTTON 4 to change METER ITEM SOURCE (meter color will change to the channel's color). LONG PRESS RIGHT TOGGLE AGAIN to revert to regular menus.

NOTE that depending on the source, some items may not be available (eg: DC volts from digital CH C or D) >> ALSO NOTE that while you can change from large to small to no meters without affecting custom meter settings, CHANGING METER PAGES will RESET custom settings. IF IT IS DESIRED TO SAVE CUSTOM METER SETTINGS, save the settings to one of the additional CONFIG FILES. Meter settings of the displayed page will be restored upon reload, along with all other settings.

METER ITEMS: Customizable items that meters can be set to display: (color indicates source channel selected) NOTE that only time based items can be displayed with digital channels.

Voltage ranges displayed: mV, V Frequency ranges displayed: mH, Hz, Khz, Mhz Time ranges displayed: nS, uS, mS, S

Vbt = Battery voltage FPS = Frames per second.

Display refresh rate (note that at the slower time bases, frame rates will be faster than data refresh rates) Vdc = Average voltage of the displayed waveform RMS= Effective DC equivalent of an AC waveform (note that this includes any DC offset present with the waveform) Frq = Frequency of displayed wave (Needs to have at least one complete period within display, starting with a positive alternation) Max = Maximum voltage attained by waveform from entire buffer.

Min = Minimum voltage attained by waveform from entire buffer.

Vpp = Maximum peak to peak waveform voltage Per = Time span of one complete measured waveform, averaged from all waves in entire buffer Dut = % duty cycle, time waveform was above mid point in relation to when it was below TH = Time waveform was above mid point within one complete measured waveform, averaged from all waves in entire buffer.

TL = Time waveform was below mid point within one complete measured waveform, averaged from all waves in entire buffer.

Cursor-restricted measurements

To toggle mode HOLD BUTTON 4 until notification appears. An "X" will be displayed in status area with mode active. All meter readings will be restricted to between the time cursors T1 and T2. For frequency and time meters, at least one positive alternation followed by a negative alternation must be enclosed for the time meters to register.

The "trigger point" in calculating time measurements will be half way of the peak to peak value of the enclosed signal. This can be changed by using the V1 and V2 cursors. The V1 cursor, if brought down below the middle of the signal, will lower the trigger point as it goes down below it.

Similarly, the V2 cursor if brought up above the trigger point, will raise it. The V2 cursor has precedence over V1, so if V1 is brought down below V2, V2 if brought up, will define the trigger point for measurements.

T1 and T2 can independently be disabled while in this mode by bringing T1 all the way to the left of the screen, or T2 all the way to the right of the screen.

With T2 all the way right, all data to the end of the buffer is included, if T1 is all the way left all data from the start of the pre-trigger section (150 samples before the trigger point indicator vernier) will be included. Works whether in RUN or HOLD, and at all Xposition settings in both buffer modes (long or single frame).

In DETECTOR mode, cursor select mode will show a frequency display at the bottom of the screen (see using advanced gen functions)

Min/max hold and alarms

HOLD MODE allows the HIGHEST/LOWEST displayed values to be recorded and saved. To toggle, HOLD RIGHT TOGGLE CENTER BUTTON until notification appears. "H" will be displayed in the status area when mode is active. MIN and MAX items in meters will show in WHITE. Values will show in SOURCE COLOR. Press HOLD, then RUN to RESET.

Changing any parameter which is related to Y display will also reset function (eg: Y range, Y pos, etc). Changing buffer size will also reset function. ALARMS are also ENABLED in this mode. When waveform either goes ABOVE V1 or BELOW V2, buzzer will sound as long as the waveform is either above or below, and for approx 1 second after.

DISABLE ALARMS: Move V1 cursor to the very top of the screen to disable it, and disable V2 by moving it to the very bottom of the screen.

Cursors

Shift menu selection with RIGHT TOGGLE until V1/V2/DELTA-V or T1/T2 are selected. Change items with BUTTON 4. Shift cursor positions with LEFT TOGGLE. With DELTA-V selected, SOURCE CHANNEL of the display can be changed with LEFT TOGGLE.

NOTE THAT CURSORS HAVE ADDITIONAL FUNCTIONS: -IN TIME BASED TRIGGERING TYPES: The difference in position between T1 and T2 determine the reference to whether a low/high pulse is either shorter or longer than to trigger on. -IN FFT MODE: T1 can be used to indicate frequency. V2 can be used to indicate level in decibels. -IN CURSOR SELECT METER MODE: T1 and T2 can select which part of the waveform to measure. V1 and V2 can be used to select which part of a waveform the frequency and time meters respond to. -IN SERIAL DECODE MODE: The difference between T1 and T2 sets the baud rate.

In manual mode shifting either will shift both together so start position can be changed without changing baud rate.

TRIGGER LEVEL CURSORS will show with a "T" on left side of screen in color of the selected channel. Adjust while in manual mode with trigg menu item LEV with LEFT TOGGLE. TRIGGER POINT CURSOR is an orange vertical cursor that shows the exact point when the device triggers on the signal. It's position can be shifted from the center of the screen towards the left with the LEFT TOGGLE while XPOS is selected.

Cursor readout

Position values of various cursors can be displayed at left top of screen. Display will show when menu is on either V vernier, T vernier, Xpos (will show T verniers) or trigger level if in manual trigger mode. When menu is on T cursor adjust or Xpos, display will show T1 or T2 time from trigger point.

With untriggered modes display will show T1 or T2 time from start at left of screen (mainly useful in chart mode). Chart mode timebases slower than 1S/div will display in (hours) (minutes) Seconds.

Display will show a negative number if T1 or T2 is positioned to the left of the trigger point, indicating time before trigger. V cursors will show position of cursor from Ypos waveform centerline for selected channel.

If meters are on, display will stay on, otherwise display will show for a few seconds after selecting or adjusting and then shut off. Cursor display is disabled if FFT is engaged to prevent overwriting FFT values. Also, in persistence mode display will "persist" along with waveform until updated or reset.

Display can be disabled, with menu on T vernier or V vernier, select delta time/freq with button 4 (T1 > T2 > delta time/freq) for T cursors or channel select for V cursors (V1 > V2 > chan select). Press center left toggle to toggle display enable/disable. T and V displays can be enabled/disabled independently.

Notification will show status when toggling. Save with boot config if desired, otherwise default will be both enabled. Trigger cursor will be enabled/disabled along with V cursors.

Working with files

SHIFT MENU ALL THE WAY TO THE RIGHT using RIGHT TOGGLE to access the file menu. Toggle can be just held right until it stops. With BUTTON 4 select LOAD/SAVE > FILE# > FILE TYPE (BMP, DAT, BUF, CSV, CFG, ARB, UAR, ROM) > FILE EDIT or FILE LIST CHANGE ITEMS with LEFT TOGGLE. EXECUTE file operation by PRESSING LEFT TOGGLE CENTER BUTTON

Select FILE TYPE first before engaging FILE EDIT or FILE LIST. If file LOAD is selected, file LIST will be displayed. If file SAVE is selected, EDIT BOX will be displayed. Default numbered files can be alternatively be saved/recalled using numbers selected in FILE# sub-menu.

With LIST displayed, use LEFT TOGGLE to move blinking selection up down list of filenames. Use RIGHT TOGGLE to move selection one page up/down at a time if there are a large number of entries. SHORT PRESS LEFT TOGGLE CENTER BUTTON to load selected file. Last loaded file will remain highlighted as a reference until file type is changed.

Button 3 can be used to load config files and exit file menu. Config files loaded from list will have name displayed on left side of screen. Characters not in the Quad's character table (if renamed externally) will be displayed as a dot (eg: tidle "~" used by Windows to truncate long file names, followed by a number).

BOOT CONFIG 0 will NOT be displayed in list, as this is a WPT file with a unique name.

DELETE FILE: With LIST displayed, delete selected (blinking) file by LONG pressing RIGHT TOGGLE CENTER. Press again to confirm delete, any other button to cancel.

With EDIT BOX displayed, change selected character with LEFT TOGGLE. Available characters are, in sequence: MISC SYMBOLS > 9-0 > SPACE > A-Z Filenames of previously loaded BMP and BUF files will show as default in edit box if file type has not been changed, otherwise, spaces will show as default.

LAST LOADED CONFIG FILE FROM LIST or LAST EDITED & SAVED CONFIG will remain as default when engaging edit CFG, even if file type has been changed. RESET selected character to space with SHORT press RIGHT TOGGLE CENTER BUTTON, reset ALL characters with LONG press RIGHT TOGGLE CENTER BUTTON. SPACES can be left in filename.

MID-NAME spaces will be replaced with underscores, all other spaces will be removed.

RIGHT TOGGLE can escape from both edit box and file list. To keep selected position on either edit or file list after leaving functions, first shift over to another sub-item with button 4. When re-engaging editing or list, previous selection will be restored, unless file type has been changed.

Supported file types

SAVE BMP for a screenshot of the device. Can be exported via USB or loaded back into the device to display. -LOAD BMP to view saved screenshots. -SAVE DAT files save the screen trace buffer from all 4 channels. Only what is visible on the screen is saved. -LOAD DAT loads saved screen buffer traces into CHANNEL D <REC_A> <REC_B> <REC_C> <REC_D> buffers. Selecting these from the CHANNEL D MENU SOURCE will display one of the 4 channels.

SAVE BUF saves the entire buffer (1 screen in single screen buffer mode or ~10 screens in large buffer mode) for all 4 channels along with the relevant time base, channel range, and buffer type information. -LOAD BUF loads a saved buffer file and puts the device in HOLD for displaying.

If saved from a large buffer, XPOS can be used to scroll the window through the buffer. Engaging RUN will exit display and return to previous settings. Serial, SPI and I2C modes can be used with BUF file loads but the timebase should not be changed.

SAVE CSV saves an exportable file to load in external spreadsheets and waveform display utilities. The first field contains a time reference for each sample, starting from the trigger point. Samples before the trigger point with show as negative numbers, indicating the time before trigger.

In untriggered modes (eg: chart mode), start point will be at first sample. If digital channels are OFF, channels A & B will be in the next 2 fields with a scaled voltage value referenced to the 0 baseline.

If an OS buffer mode is active, 2 sets or fields will be present for each analog channel, the first representing the high values and the second the low values. Both can be superimposed to represent the oversampling display.

NOTE that at 2uS/div and faster for 8x oversampling (red buffer indicator) the buffer mode reverts to a NON oversampling mode because of buffer size limitation.

With FULL SPEED oversampling (blue buffer indicator, available only with HW2.81 and V1.1 FPGA) the buffer reverts to a NON oversampling mode at timebases of 0.2uS and 0.1uS/div, since sampling is already at full speed for each sample.

If any digital channel is ON, all 4 channels are represented as they appear on the device to display timing relationships, with numbers representing ADC steps or vert pixels.

LOAD ARB loads a comma delimited text file with a sequence of numbers defining output of the arbitrary wave generator function. Output type selected must be ARBTR before loading the file for this to have any effect.

LOAD UAR loads a file of any format for generator serial transmission. File can be of any length, but only the first 4096 bytes will be transmitted. Output type selected must be UART for this to have any effect. Transmission starts after this file is loaded.

SAVE CFG saves an alternate config file that can be recalled later to restore the saved settings.

Up to 9 of these can be saved if using the default numbering method, in addition to the default boot-up config. A number in the status area shows the presently loaded config. #0 will always be the default boot config (XXXX.WPT) while the additional files have a CFG extension.

Any number of additional config files can be saved if edited with the filename edit function. When these are loaded after selecting from the filename list, the name will show at the top left of the screen, just below the RUN/HOLD display. HOLD BUTTON 3 for a shortcut to SAVE CONFIG, then press BUTTON 3 AGAIN to save.

Default will save file #0; change number with left toggle to save to an alternate file or edit filename before pressing button 3 again if desired.

LOAD CFG will restore previously saved settings. LOAD CFG#0 with the file# flashing will be the default file menu setting after boot-up. This makes it easy to select alternate configs: just hold RIGHT TOGGLE right until menus stop, CHANGE CONFIG NUMBER with LEFT TOGGLE.

OR engage file list to load highlighted blinking file, then press LEFT TOGGLE CENTER BUTTON to load. File menu will stay with numbers or file highlighted, making it easy to "browse" though other configs. OR press BUTTON 3 to load and restore menu selection saved with config.

SAVE ROM: This saves a 512KB binary image of the complete ROM, including bootloader, licence, SYS, all apps, FPGA and logo in .BIN format for restoring device via internal header if ever necessary. Filename is set to ROM_IMG.BIN and cannot be changed.

Make sure enough space is available on drive before using this, or saving ANY TYPE OF FILE if the drive is ALMOST FULL, specially on 2MB devices. If there isn't, the system's file access functions leave the failed entry behind and uses up the remaining space.

Deleting the entry may not restore the lost space, only after repairing the volume with a utility such as CHKDSK (from the command line in Windows) or reformatting can the lost space be recovered.

SECTION 06

FFT, spectrograph and MAP

Frequency-domain analysis and time-varying spectral views.

FFT spectrum analyzer

FFT mode is part of CHANNEL D. It can quickly be accessed by moving left toggle left while in CH D menu, if set to OFF. You can scroll backwards: hFFT channel B < sFFTchannel B < hFFTchannel A < sFFTchannel A < OFF 2 window modes are available: Hann (h) or Summing (s).

The SUMMING MODE is not actually a window, but a function that sums any detected peak with it's neighbors. This provides an absolutely flat frequency response, at the expense of the two frequencies above and below the peak. For most measurements this won't be noticed, and is the preferable mode to use.

When displaying signals of constantly varying frequency (such as music), the "every other frequency missing" notches may be apparent. In such cases it may be preferable to use the Hann window. It should be noted that in this mode, ALL frequencies CAN be displayed. Only where there is a peak are the neighboring frequencies restricted.

TO MINIMIZE ALIASING buffer mode can be set to either AVERAGING or 8x OVERSAMPLING. In either case, AVERAGING WILL BE USED. There is no advantage to using the full or large buffer mode, this will only slow the frame rate down.

FULL SPEED OS MODE (available only with 2.81 and up devices) can give poor results, particularly towards the right half of the screen where the waves are clustered close together: other buffer modes are generally cleaner and more accurate.

FFT magnification

in CH D menu, while in one of the 4 FFT sources, press BUTTON 4 to display MAG modes in notification area. 2 modes are available: AUTO and T1 mode. Toggle between the two with RIGHT TOGGLE CENTER BUTTON. Engage either mode by pressing LEFT TOGGLE CENTER BUTTON. Press again to leave mode.

When this is engaged, waveform trace and regular meters will be disabled, as their display would be irrelevant. Higher resolution frequency meters will be displayed instead. Wave trace can be turned back on manually while in this mode but regular meters will be locked out. AUTO MODE: Used to lock on to and display a triggered signal.

Adjust for a stable waveform display with a readable frequency with the spectrum peak near center. If 2 or more frequencies are interfering with each other preventing a stable waveform and frequency display, use T1 mode instead. An unstable display before engaging will result in inaccurate readings in auto mode when magnified.

With "AUTO MAG: 0x" displaying in notification area, press left toggle center button. This engages the MAG mode, centers the peak and engages higher resolution meters. Notification area will show MAG: 1x.

This "base level" represents an approx 5 fold expansion over the regular display, depending on where the peak was positioned in the regular display before engaging. An additional 2000x is available in 2-5-10x steps by using the LEFT TOGGLE. The program will attempt to re-center the peak when magnifications levels are changed.

Use the RIGHT TOGGLE to SHIFT the WINDOW left or right to position peak if necessary: If the window is shifted so far with the right toggle that the peak goes beyond the right or left side, reading will be lost: shift the window back to restore.

NOTE that if the waveform peak itself however drifts beyond the sides, it will "fold back" with an incorrect reading, keeping the peak within the window with the right toggle will prevent this. Continued on next page

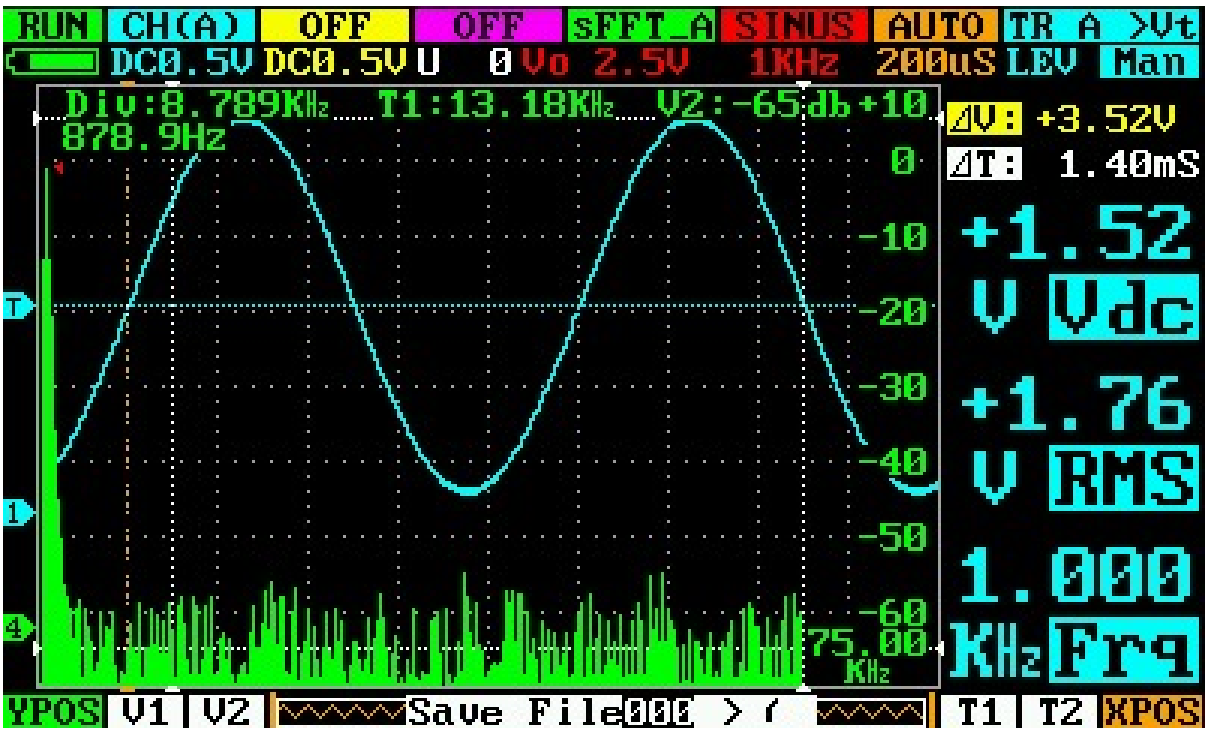


FIG. 09 FFT spectrum: 1 kHz signal, logarithmic frequency display.

At right/bottom of the display, start and end meters will show the frequency at the left and the right side of the spectrum window. At higher magnifications, meters will show greater resolution and accuracy, however, the screen refresh rate will slow down, limited to about 3 - 4 or so seconds per frame as the slowest for lower frequencies.

If for any reason the display is suspected to have shifted beyond the window limit, because the frequency has drifted or the window has been shifted too far, press LEFT TOGGLE CENTER BUTTON TWICE TO RE-ALIGN. The second press will return to the previous magnification level. Shift window back to center if necessary with the right toggle.

Magnification level with mode engaged will remain until changed, T1 mode is engaged, FFT is disabled or the device is restarted. It is not saved in configs. T1 MODE: Used to lock on to the frequency indicated by the T1 cursor BEFORE engaging mode. First adjust for the desired frequency with the timebase and T1 cursor.

The T1 CURSOR can be SHIFTED with the LEFT TOGGLE - BEFORE engaging mode. Then with T1 MODE MAG= 0 displayed, press LEFT TOGGLE CENTER BUTTON.

Default magnification is x1, since as magnification is increased with the LEFT TOGGLE, the frequency likely will have to be SHIFTED MANUALLY with the RIGHT TOGGLE, to stay aligned with the T1 cursor by moving the window. Shifting the window to the LEFT will INCREASE the frequency at the T1 cursor, shifting it to the RIGHT will DECREASE it.

The T1 cursor can be used to adjust to the exact frequency, but it will be reset if the window is shifted in the highest magnification level selected.

For example: If magnification is increased all the way to x2000 and the T1 cursor is shifted over to some peak to determine it's exact frequency, shifting the window to move the display will reset the cursor to it's mid-screen position.

However, backing off to x1000, x500, etc, everything will remain in place, and shifting the window at these lower magnification levels will not affect the T1 position, only if magnification is brought back again to x2000 (or any other max level attained) will shifting the window reset the cursor.

Resetting by pressing the left toggle center button twice while the notification is showing T1 MODE will reset display to x1.

IN BOTH MAGNIFICATION MODES: LONG PRESS BUTTON 4 shortcut to time/div is changed to a SHORTCUT to TOGGLE BETWEEN MAG MODE AND T1 ADJUST WHILE THE MODES ARE ENGAGED, OR T1 MODE SELECTED,

otherwise, if the MAG sub menu is NOT selected, the shortcut works normally. This makes it easy to adjust the cursor in T1 mode.

Settings such as timebase, meters and cursor position changed while in these modes will be restored upon leaving. ALSO NOTE that FREQUENCY ACCURACY will depend on properly compensating the CPU clock with an external standard with the PPM clock adjust function.

PPM adjust can now also be accessed without entering calibration, while displaying the ADCoffset control, use the left toggle center button to toggle between the two. >> To determine whether additional waveform peaks are sidebands associated with a frequency, shift the window with the right toggle.

If peaks are part of the main waveform, they will shift with it. If they are just background peaks not part of the window's range, they will move away to random positions. Often, background harmonics of the main signal can be hidden this way, either pushed over to the sides or behind the main peak.

FFT GAIN can be adjusted by a factor of 42db. Dynamic range is such that even with +42db gain setting, a full scale waveform will not overload. TO ADJUST FFT GAIN, while in the CH D menu, and in any FFT/SPECTROGRAPH mode, press BUTTON 4 (source > magnification mode > gain adjust): A gain display will show in the notification area.

Use left toggle to change. Selecting towards the left will show AUTO. In this position, gain setting will automatically change in relation to the signal level. For constantly changing waveforms (music for example) or to gain extra resolution with large amplitude signals, switch over to manual settings, (0 to +42db in 6db steps).

LOG AMPLITUDE SCALING can be selected in notification area gain menu LOG > AUTO > 0db > +6db > .... > +42db In FFT mode, V2 CURSOR functions as a level reference (shown at top of screen). Value for this as well as the db scale on the right will change with different gain settings.

In a similar way, T1 cursor can be shifted to indicate frequency. NOTE that the frequency displayed on peaks is not the frequency of the signal but the "bin" center frequency. Bins at the far left carry a RELATIVELY wide range of frequencies and may not show the exact frequency of the signal.

Accuracy will improve for peaks away from the left edge. PERSISTENCE MODE: If this is engaged while in FFT mode, will act as a "peak hold", as all display frames will stay on screen. Reset with any button, or freeze with HOLD and press again to reset.

Note that in this mode, the "floating" peak frequency display and red indicators will be disabled, as any reference to the actual peak will be lost. TO DISABLE PEAK FREQUENCY READOUT AND INDICATORS: Shut meters off. Readout will only show with meters on.

This may be desirable for example when displaying music, as the indicator will be doing much flinging around.

Spectrograph

THIS IS ALSO PART OF CH D: In DIGITAL CHANNEL D MENU, select SPEC. Either channel A or channel B MUST be turned on, and selects the input for this mode. If BOTH are ON, CH A has precedence and will be displayed. SPEC MODE displays the FFT frequency spectrum on vertical rising lines, one for each frame.

The COLOR at any given point defines the LEVEL, from dark blue > cyan > white > red as the amplitude rises, and the HEIGHT defines the FREQUENCY.

TO DISPLAY ENVELOPE: SHORT press RIGHT TOGGLE CENTER BUTTON to cycle SPEC > SPEC WITH ENVELOPE > ENVELOPE ONLY. A MOVING CURSOR (short yellow line) will display at top of screen to show scan position, frequency scale at the left and seconds "ticks" will be displayed in all but the slowest time bases.

ADJUST GAIN and scaling in the same way as with FFT GAIN adjust. Does not affect envelope level.

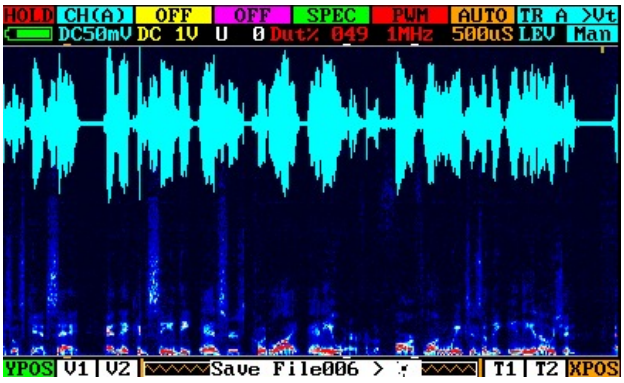


FIG. 06 Spectrograph: voice print of a person speaking.

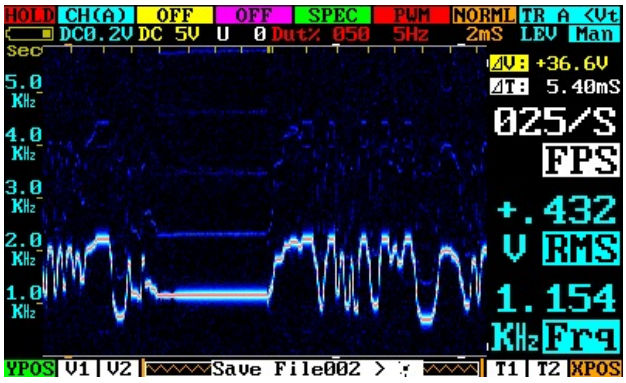


FIG. 10 Spectrograph: frequency scale reference.

MAP display

CHANGE CHANNEL D MENU TO MAP. Either channel A or channel B must be turned on, and selects the input for this mode. If both are ON CH A has precedence and will be displayed.

MAP MODE displays normal oscilloscope waveform level as a changing COLOR, each frame on a single horizontal line, so therefore has the ability to display 200 simultaneous frames. COLOR varies according to signal position on screen, with bottom of screen showing as black or dark blue, middle as cyan, and top as white or red.

Note that adjusting Y POSITION CHANGES THE BACKGROUND, if no signal is displayed or changes the signal mid-point color if a signal is displayed.

WAVE CALIBRATION ENABLE/DISABLE Toggle WAVE CALIBRATION ON > OFF with LONG PRESS LEFT TOGGLE CENTER BUTTON WHILE METERS ARE OFF. "C" for calibrated or "U" for uncalibrated will display in the status area. Turning calibration OFF can be useful to preserve waveform quality.

Calibration compensation is accomplished by introducing "steps" along the waveform, increasing or decreasing amplitude. Under some conditions these can be visible and detract from showing an accurate display of the monitored waveform. Has NO EFFECT on METERS, these show calibrated values at all times.

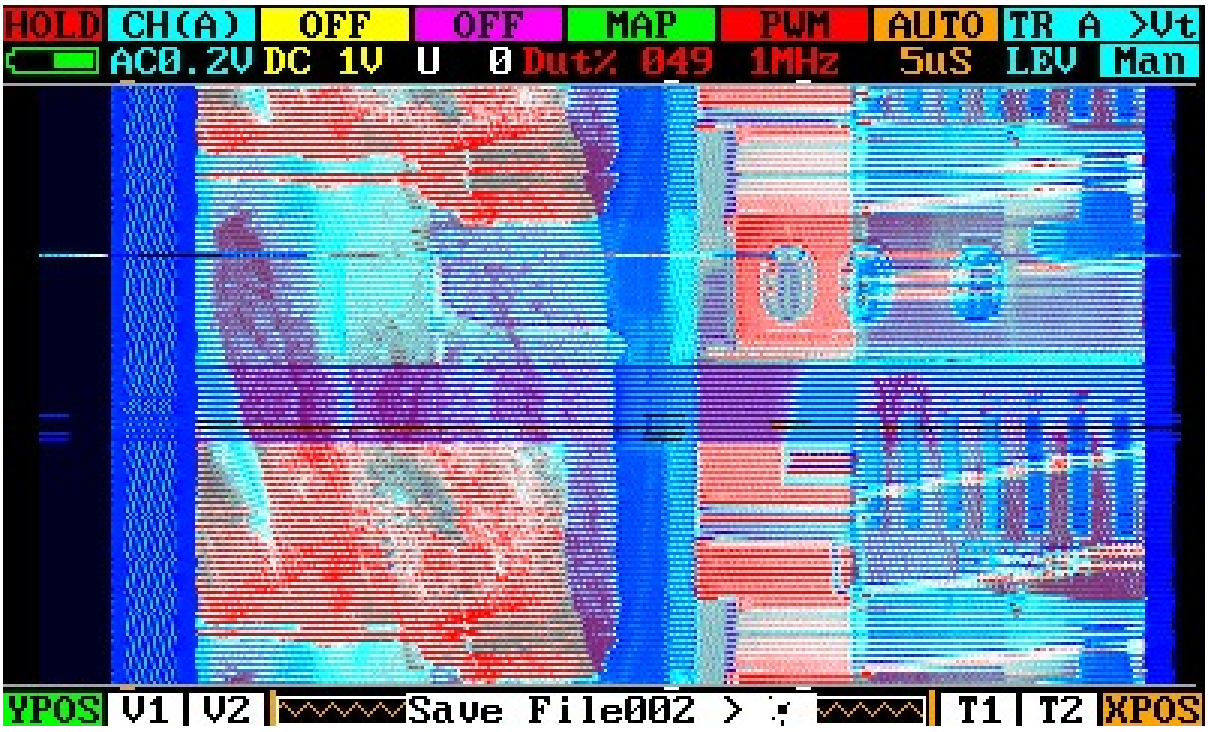


FIG. 07 MAP mode: visualizing a video frame.

SECTION 07

Waveform generator and advanced output

Continuous signals, PWM, arbitrary waveforms, sweep, burst, noise and serial output.

Advanced waveform generator

Sweep and burst modes are available for all wave types except noise and arbitrary.

SWEEP AND TONE BURST GENERATOR: While in OUTPUT MENU, FREQ sub menu, SHORT press LEFT TOGGLE CENTER BUTTON to display special generator functions. SHORT press RIGHT TOGGLE CENTER BUTTON is then used to cycle through FREQ ADJUST > FREQUENCY SWEEP PERIOD > TONE BURST RATE > TONE BURST DURATION > TONE BURST FREQUENCY > back again to FREQ ADJUST. Relevant value will be displayed in notification area. Change values with LEFT TOGGLE.

TONE BURST RATE and TONE BURST DURATION are shown in same display, use LONG press RIGHT TOGGLE CENTER BUTTON to go backwards from BURST FREQ to BURST DURATION to TONE BURST RATE if necessary.

While in sweep mode or burst mode, output display will show in WHITE. In continuous, arbitrary or OFF mode, display will show in RED.

All modes will remain AS SET if menu is shifted away and will be saved with configuration. When not using generator, shifting wave type to OFF will allow all resources to be available to scope function and provide best performance.

TRIGGERING on sweep and bursts can be facilitated by selecting "GEN" mode in triggering type menu. In this mode, triggering is set to trigger after the time span between sweeps or bursts, on the first positive transition if the trigger cursor is above the wave center, or the first negative transition if it's below.

NORMAL SWEEP MODES should be used, as auto mode can time out between sweeps/bursts and cause instability. AC COUPLING should also be used if DC from the generator is present at the input, as it will prevent the program to find the wave center.

ALSO, in this mode, auto trig is restricted to 1/4 and 3/4 level of wave as a transition may not be detected at the half way point. MAN trig level gives best results since spaces between sweeps/bursts can cause auto trigger instability.

GEN menu selection COLOR will change from channel color to GRAY if the combinations of settings (most likely timebase) or hardware limitations (eg: max time between sweeps > 4096 samples, limited by 12 bit variable in FPGA) can't guaranty triggering even if trigger level is properly set. In such a case, triggering may work ok but may not as certain settings are changed. In such a case, change timebase until GEN changes back to channel color.

RANDOM NOISE MODE: Shift wave type to NOISE. Change bandwidth with FREQ sub menu. Noise mode is not adjustable as to output level. At 20Khz and below, noise is filtered with a 10X oversampling filter, below 1Khz with a 100X oversampling filter. Above 20Khz no filter is used and the generator will output raw digital samples.

PULSE MODE: Shift mode to PWM continuous (shift output function using left/right toggle center buttons if necessary to move out of sweep or burst modes). While in OUTPUT MENU change sub menu to %DUTY using BUTTON 4.

Adjust %DUTY with LEFT TOGGLE to the approximate value desired (for very short pulses this can be set to either 0% for positive pulses or 100% for negative pulses). While in %DUTY sub menu SHORT press LEFT TOGGLE CENTER BUTTON to access PULSE LENGTH display in notification area. This also engages pulse mode.

Note that engagement of this mode is NOT saved with config files. Adjust pulse length with LEFT TOGGLE. For large changes, shift back to %DUTY with LEFT TOGGLE CENTER BUTTON, then back again to set to desired length.

If %duty is less than 50%, pulse length will show as +value, if more than 50% length will show as a negative pulse length value, decreasing as the setting approaches 100%. -While in PULSE MODE, repetition rate (frequency) can be adjusted by

shifting sub menu to FREQ and changing range or by engaging the VARIABLE FREQUENCY DISPLAY in the notification area.

Pulse length will not be affected by changing frequency. If very small values are used and the frequency is shifted to lower values, length might then increase if the set values are less than the minimum values supported.

Also, while shifting ranges, some set length values may shift very slightly longer/shorter as different available division ratios are engaged.

ARBITRARY WAVE MODE: First shift wave type to "ARBT". Then access file menu, select LOAD, ARB type and load desired file. Change frequency range and adjust frequency in the normal way. ONLY FREQ sub menu and FREQ ADJUST mode with display in the notification area are supported in this function.

Wave will stay in memory until a different wave type is selected. Maximum frequency range available will depend on the number of samples used. -GENERATING ARBITRARY WAVE FILES: This is a plain text file with a sequence of numbers ranging from 0 (max neg value) to 4095 (max pos value).

Commas are used to separate the numbers as in a CSV (comma separated values) file. They can be generated manually or with any program that can create these type of files. Compilers and interpreters for example can easily write a math formula or algorithm as a sequence of numbers to a file.

The program will ignore any whitespace or other characters other than numerals or commas. 3 simple rules apply: 1) First value MUST define how many samples are used. This can be anything from 2 to 720 samples. 2) There MUST be at least as many following numbers describing the wave as the first sample number indicates.

If there are not, the program will generate an error upon loading. If more are present the program will simply ignore the surplus. 3) The last value MUST be followed by a comma. Programs that generate these types of files typically do NOT add a comma after the last field, so this may have to be added manually with a text editor.

Alternatively, an extra value could just be added.

FOR EXAMPLE: The simplest waveform would be a square wave and could written like this: 2,0,4095, and would be a full amplitude square wave with a frequency adjustable from 1Hz to 900Khz. SAVE or RENAME the file with a filename in the format of DATAxxx.ARB where xxx can be anything from 000 to 255.

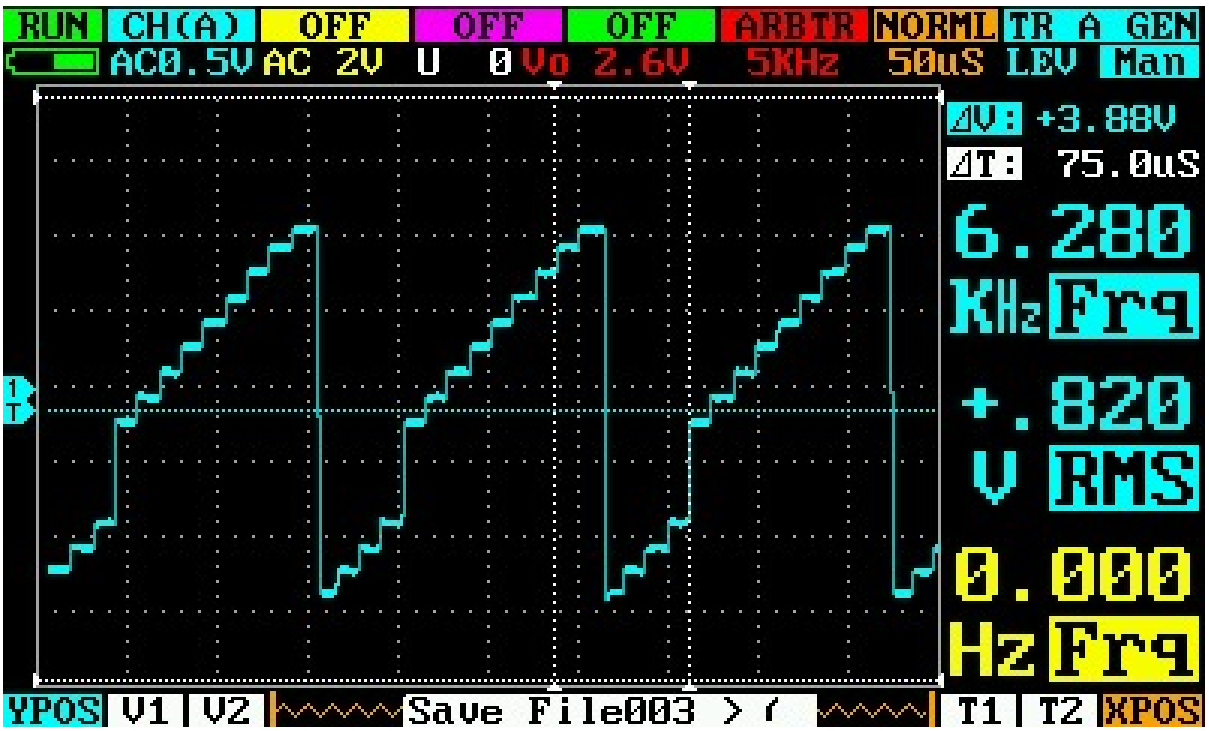


FIG. 16 Arbitrary waveform generator output.

SERIAL CODE GENERATOR: Shift wave type to "UART". Change menu item to "FREQ". Notification area will show parameters to adjust. Use SHORT PRESS RIGHT TOGGLE to change blinking cursor to select BAUD RATE, FORMAT, STOP BITS and CONTINUOUS modes. Adjust BAUD RATE from ~1100 to 4,500,000 with LEFT TOGGLE.

Use SHORT PRESS LEFT TOGGLE to change between FAST and FINE adjust rates. Change FORMAT 7Pe > 7Po > 8N > 8Pe > 8Po > 9N with LEFT TOGGLE, where 7/8/9= Number of bits (including parity), Pe= Even Parity, Po= Odd Parity and N= No parity. Select STOP BITS: 1 or 2 bits (1S or 2S) with LEFT TOGGLE.

Select SINGLE TRANSMISSION or CONTINUOUS (1 or C): In SINGLE mode data is output once after a file is loaded, then the line is idled (high logic). In CONTINUOUS mode, data from the file is continuously resent, without any pauses, breaks or other interruptions.

INITIATE SERIAL TRANSMISSION by LOADING a DATAxxx.UAR file from the FILE MENU while the generator is in UART mode. UAR file can be of any format. The program will parse it out and transmit it as a sequence of single bytes. File size is limited to 4096 BYTES, if the file is longer only the first 4096 bytes will be transmitted.

STOP TRANSMISSION (if in continuous mode) by changing ANY of the adjustment parameters (eg: change "C" to "1")

Unavailable function combinations

Some functions or modes have been disabled under certain settings, either because the combination made no sense, the hardware did not support them or because of programming issues.

Some examples are: Invert modes while in any of the 3 serial decode functions; averaging and oversampling buffers while in TrOFF, XY, detector or A & B trigger modes; digital channels with averaging or oversampling buffers or A & B trigger modes; SPEC, MAP or any of the SERIAL, SPI or I2C decode functions while in CHART mode and any but NORML time base mode with A & B trigger source.

If any function seems disabled, check settings carefully to make sure they support it.

SECTION 08

Detector and frequency-response measurement

Use the built-in sweep generator to inspect loudspeakers, filters and other frequency responses.

Detector mode

USING DETECTOR MODE TO DISPLAY FREQUENCY RESPONSE: 1) Engage sine sweep mode: Shift generator menu to SINE, then with sub menu on FREQ short press left toggle center button to turn on special functions. Short press right toggle center button to shift to sweep mode (notification area will show sweep period adj).

2) Set scope sweep to NORMAL (auto mode can time out between sweeps and cause instability) 3) Set triggering to GEN 4) Set chA or chB coupling mode to DT and turn on channel. 5) Set buffer to single frame at first: Quickly scan all frequencies by shifting generator freq range. Entire sweep for range will be contained within window.

Xpos may need to be shifted to the extreme left on some of the ranges to show entire sweep. 6) With a signal from the generator fed back into chA or B, freq response curve should be visible. Adjust vert range, Y pos, and trig level if necessary. Note that AUTO trig level in GEN mode with sweep on is restricted to 3/4 and 1/4 only.

MAN works best for this function and can be used to "lock" 3/4 level after selecting it by engaging MAN with button 2 long press. 1/2 level or close to wave center should be avoided as scope needs to trigger going either above or below this point, so if trig level is too close to the center line triggering will not know which way to go.

To allow program to auto set, change frequency RANGE with output menu on FREQ. Make sure to adjust freq and not special function parameters (eg: sweep period). In other words, ADJ parameters should NOT be displayed in notification area, if necessary, short press left toggle center button to go back to FREQ.

When changing frequency range in this manner (with at least one channel on in detector mode and sweep mode with one of the analog waveforms selected) the program will automatically adjust SWEEP PERIOD and SCOPE TIMEBASE to properly display response curve. Entire frequency range can be quickly scanned this way to find relevant area of response curve.

To get BETTER RESOLUTION, shift buffer to FULL. In the same way as with single frame buffer, changing frequency range will adjust sweep period and timebase automatically for a wider displayed sweep appropriate for a full buffer. Xpos will be now be necessary to be used to view entire sweep. Note that more than 1 sweep or a partial 2nd sweep may now be visible at the right after shifting Xpos.

SWEEP PERIOD and TIME BASE can be manually adjusted, if desired, keeping in mind the program first adjusts the sweep period, then timebase to match. Therefore, sweep period can be manually adjusted without changing frequency range, but timebase will be readjusted. Finally, timebase can be then adjusted without anything else changing.

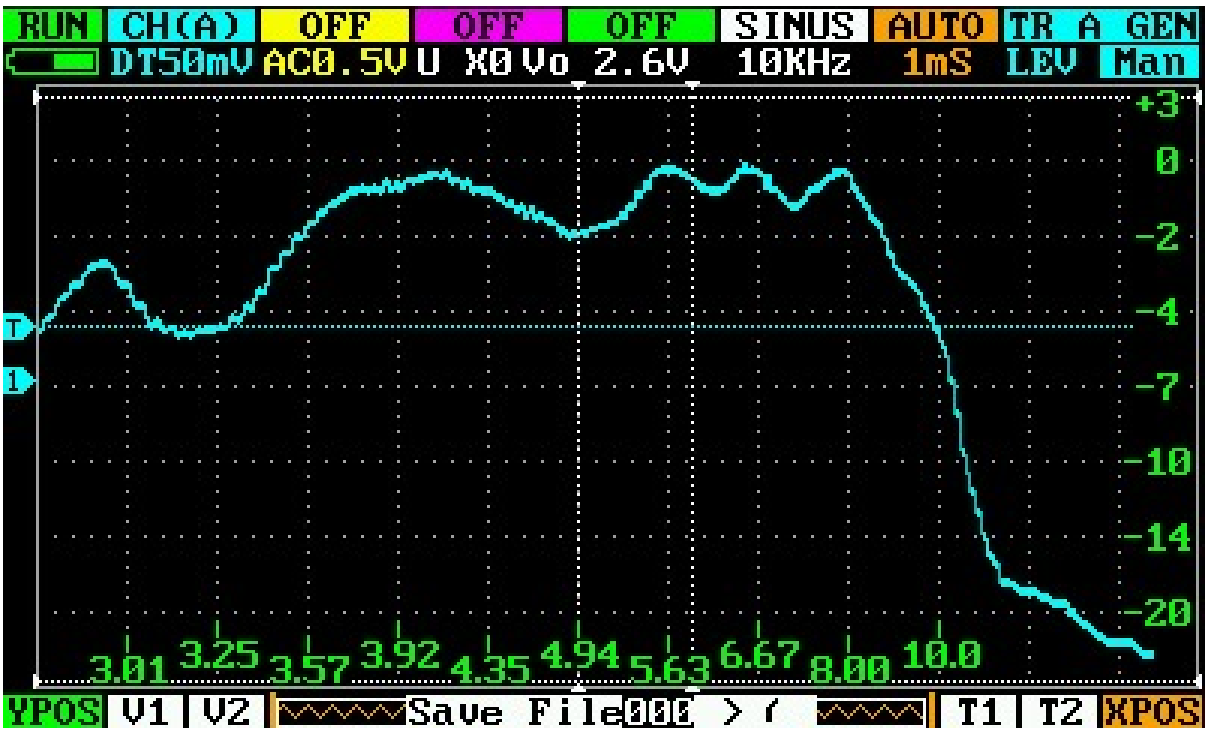


FIG. 11 Frequency response: loudspeaker, 2.5–10 kHz sweep.

To display FREQUENCY SCALE, move T1 and T2 to center, or somewhat a bit right of center with distance between the two at something like 1 to 2 divisions, wide enough to display at least on complete wave. Readings will be displayed no matter where the cursors are but accuracy will be best if they are close to the middle.

THEN HOLD BUTTON 4 DOWN to engage CURSOR RESTRICT FREQ METER. Note that meters do not have to be showing but it may be useful to have meters on for monitoring. Once freq readout is gained, the program will extrapolate a frequency scale at the bottom with a readout at each division.

ALSO NOTE that scale is estimated, particularly in single frame buffer mode, with accuracy typically within +/- 2 to 3%. For best accuracy, shift to full buffer mode and a properly expanded sweep, or position cursors around position of interest and read FREQ meter.

The distance between the two T cursors must be wide enough to contain at least one complete wave or a reading will not be obtained. -It may be advantageous to save such a configuration to one of the extra available positions to quickly recall if desired.

PWM wave type CAN be used in detector mode, but noise and aliasing artifacts will become significant at the higher frequencies. PWM inter sweep level will be shifted from 0 (as is normal for this mode) to 1/2 of the wave amplitude and duty set to 50%; however auto setting of sweep period and timebase will NOT work using PWM waveforms. Settings will have to be made manually.

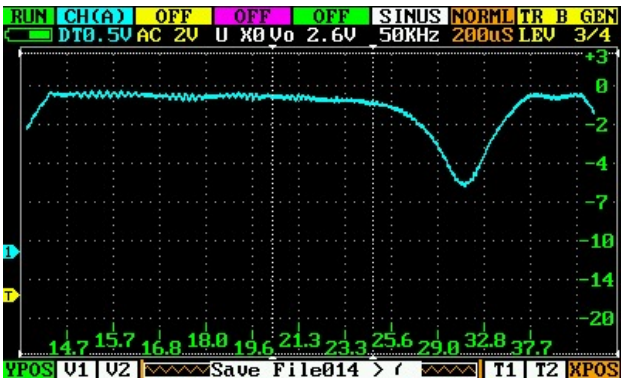


FIG. 13 Detector mode: fast notch-filter scan.

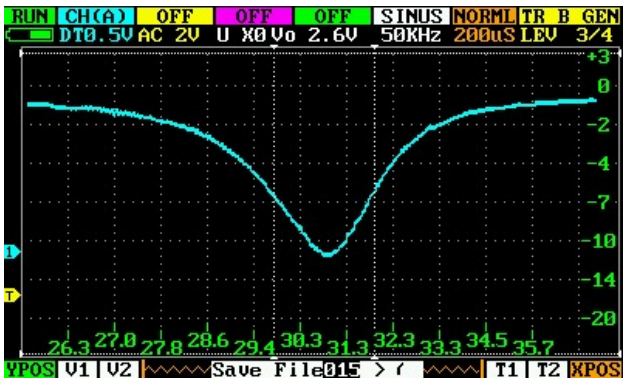


FIG. 14 Detector mode: expanded notch-filter scan for greater resolution.

SECTION 09

Serial protocols: UART, I²C and SPI

Decode captured data from the oscilloscope channels and inspect hexadecimal and ASCII values.

RS-232 / TTL serial decoding

RS-232 / TTL decoding

This is only available using analog channel A: With menu on CHA, change item to COUPLING, then using LEFT TOGGLE cycle AC > DC > DET > TL > RS and select either TL (TTL, for positive logic) or RS (RS232 for inverted logic). Coupling will be set to DC in both modes.

Notification area will show BAUD RATE, DATA LENGTH AND PARITY (N for none or P for any type). While in CHA MENU, and COUPLING blinking, short press LEFT TOGGLE CENTER BUTTON and adjust DATA size with LEFT TOGGLE: change from 5bits >>> 8bits with no parity > 5bits>>> 8bits with parity.

To return to coupling press LEFT TOGGLE CENTER BUTTON again. FFT, spectrograph and MAP modes must be OFF. Set timebase mode to NORMAL or SINGLE, buffer to FULL if more than just a few bytes are expected and other parameters to properly display waveform in top half of screen (bottom half is used to display characters/hex data).

BAUD rate RANGE is changed along with the normal scope timebase: select timebase to properly display signal, then shift menu to T1-T2 ADJUST. Baud rate is set by changing distance between T1 and T2. Cursors can be anywhere, only the distance between the two is relevant.

It can be useful though to set T1 at start of wave: for example, if baud rate is unknown, T1 can be positioned at start of waveform (start bit) and T2 moved to approximate character frame length, defined as number of data bits + 2 (start and stop bits) + 1 parity bit if present. Extra stop bits are ignored.

Once rate is within approx +/- 5% of correct value the decoder should synchronize and show a valid display.

Timebase used in relation to baud rate will determine how many characters are displayed: selecting slowest time base possible while retaining a clearly defined waveform (with T1 and T2 around 1 division apart) will allow the maximum number of characters to be displayed (max is 96).

Conversely, speeding up timebase will allow for a clearer waveform but will restrict the size of data displayed.

Continuous stream synchronization

The decoder should synchronize on a continuous stream, provided enough characters are decoded. A "continuous stream" meaning a stream interrupted at some mid-point, where the start character/byte is unknown.

This will also depend on the type of data captured; if a continuous stream of a same particular byte value is sent, for example, depending on the value, the program could confuse data bits for stop bits and improperly decode them. In such a case, first set data size, parity and baud rate.

Then manually define the first start bit: With menu on T1-T2, short press LEFT TOGGLE CENTER BUTTON to engage manual mode (notification display will change from T2-1=BAUD to T2+1) and shifting left toggle will move BOTH T1 and T2 together, keeping the baud rate as set. T1 will now define the first decoded start bit, shifting it along the data stream should show a point, along a decoded byte where synchronization is correct.

The "T2+1" display color will change to help identify when in sync, but in some conditions, such as when very few or many of the same characters are displayed causing the decoder to sync on the data rather than on start/stop bits it may be necessary to observe the data itself.

Decoding beyond 96 bytes

It is possible for the buffer under some conditions to hold up to 150 bytes of data. These can be decoded by advancing XPOS to the end of the 96 byte decoded portion, engaging manual mode and shifting the T cursors. Decoding will then start at this position and display the rest of the data.

Captured data can be quickly saved in "BUF" format: The menu can be kept on file and consecutive captures written to file with one button push. Reloading each buffer file will decode the data. Alternatively, BMP screenshots can be saved, and will show decoded data but these take longer to write.

NOTES: When selecting 9 bits for decoding, then 9th (MSB) is simply discarded. The 9 bit selection is included only to allow proper synchronization. The status of this bit can be observed from the waveform, if necessary and will be located to the left of the stop bit, which is above the right byte location arrow ">". The same holds true for the Parity bit. In the latest versions (4.4 and up), serial decoding has been extended to the fastest timebases.

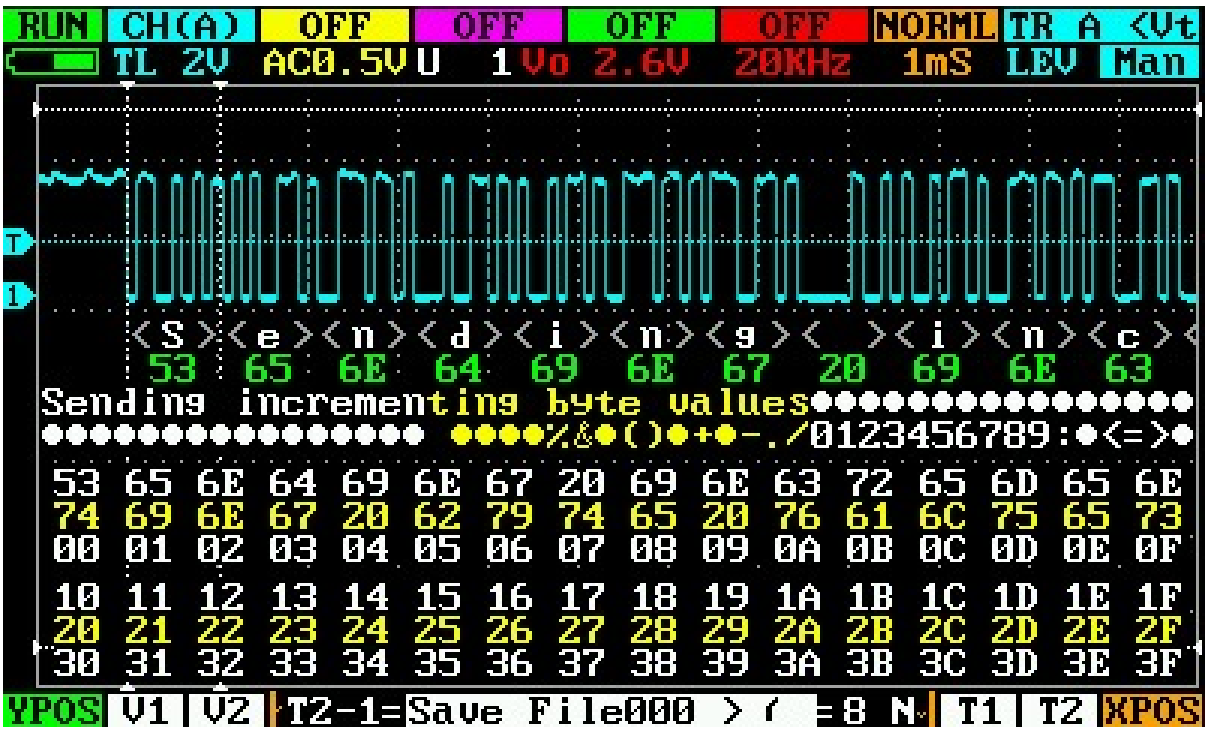


FIG. 15 RS-232 / TTL serial decoding with hex and ASCII data.

I²C decoding

Connect the DATA line to CH A and the CLOCK line to CH B. Set CH A COUPLING sub menu to I2, and ALSO CH B COUPLING sub menu to I2. Time base mode works best set to NORMAL or SINGLE. AUTO can be temporarily used to view signal if not triggered in normal mode. Setup triggering and time base to properly display CH B clock signal.

Level triggering set to neg transition usually works well, time based triggering on inter-packet spaces can also be used to advantage. Manual trig level works best as these signals tend to come in bursts. Move both channels up into the top half of the screen as bottom half is used to display buffer data.

NOTE that the triggering level must ALSO BE PROPERLY SET FOR THE DATA CHANNEL. As a reminder, the trig level arrow at the left of the screen for Ch A will remain while triggering is set to the Ch B clock. To adjust this, the trigger source needs to be set to CH A and adjusted in the normal manner.

This sets the point at which the data is sensed changing HI/LOW and can be fine tuned to eliminate the effect of noise, ringing or other waveform distortions if necessary (for example a drooping positive transition on the signal lines from weak pull-up resistors can be compensated for by bringing the trig levels down towards the base line).

Bytes displayed with first clock sequence or restart sequence will be the SLAVE ADDRESS and are displayed in PURPLE, with the associated ASCII display showing "ad". 10 bit addresses occupy 2 bytes: the first will be the 2 most significant bits of the address in byte "XX" format (0 to 3) while the second contains the remaining 8 bits.

Bytes not colored purple are data. The byte location frame indicators "< >" show the direction of data flow, with the right arrow highlighted indicating master to slave transfers and a left highlighted arrow indicating a slave to master transfer.

The color of the highlighted arrow further signifies the ACK (GREEN) or NACK (RED) bit associated with each byte.

Bottom of screen displays the content of the buffer in ASCII and HEX format.

SPI decoding

NOTE that to if using Ch C as clock, devices prior to HW V2.72 should have protection diodes at the digital inputs removed as these severely limit the bandwidth and timing, and will cause decoding to fail at all but the lowest frequencies.

Connect the 2 data lines (MOSI and MISO) to CH A and CH B. The clock signal is normally connected to CH C, but can also be connected to CH B, in which case only one of the two data channels will be displayed. If CH C is turned OFF, the program will use CH B as a clock.

In SPI mode, the word and ASCII waveform location indicators sit just below the 0 baseline of the channel, and move up/down with the waveform as the Y position is changed. CH D can also be connected to the relevant slave select line. Triggering can be set to either clock (Ch C or Ch B) or slave select, if connected.

The BUFFER CHART displays both data channels and can overwrite the entire display, depending on the amount of data, and is toggled ON/OFF from the CH B COUPLING ITEM with the LEFT TOGGLE CENTER button.

If word length has been set MANUALLY to either 7 or 8 bits, ASCII data will be displayed on the chart in addition to HEX, otherwise only HEX data will show. Data is displayed in relation to the waveform positions, if Ch B waveform is above the Ch A waveform, top display is of Ch B, bottom Ch A.

If Ch A was above Ch B then Ch A data will be above.

NOTE THAT TRIGGERING LEVELS MUST BE PROPERLY SET ON BOTH ANALOG CHANNELS. Trig level arrows at the left of the display will be displayed as a reminder. Change by selecting with trig source and adjusting in the normal way. Trig level defines the HI/LOW change point for the clock/data pulses and can be moved up/down the waveform to eliminate noise, ringing, etc.

Set BOTH CH A AND CH B COUPLING MODES TO SP. Set trig mode on MANUAL and time base/trig controls to properly display the clock signal. The SPI SETUP MENU is accessed by selecting the CH A COUPLING sub menu. Relevant parameters to be adjusted will be displayed in the notification area.

Press LEFT TOGGLE center button to shift cursor to the notification area. Change items to be adjusted by pressing the RIGHT TOGGLE center button. The CHART DISPLAY is toggled by shifting to CH B COUPLING menu while set to SP and pressing LEFT TOGGLE button.

Note that the chart display text overwrites the waveform and will not show unless data is captured.

INFORMATION ABOUT AN SPI TRANSMISSION MUST BE KNOWN AND ENTERED INTO THE DEVICE IN ORDER TO PROPERLY DECODE. NOTE that any of these parameters can be adjusted AFTER a properly synchronized signal is displayed or paused, if necessary, while observing the waveform and decoded data. All settings except for the chart enable toggle are saved in a configuration file.

Adjust MODE: Clock polarity and phase are indicated with a pair of binary digits, the first digit indicates polarity, with "0" having a low base, with clock pulses going positive, while "1" indicates a high base or rest position with clock pulses going negative. The second binary digit adjusts whether data is read from the leading clock edge "0" or trailing clock edge "1". Select the 4 possible combinations with the left toggle.

Adjust DATA BIT DIRECTION: Press CENTER RIGHT TOGGLE to shift cursor to ">" just to the right of the mode indicator bits. Change with left toggle. ">" indicates a MSB or most significant bit first transmission while "<" indicates a LSB or least significant bit first type.

WORD LENGTH can be specified in one of 3 ways: 1) Manually 2) AUTO or 3) Reset after space, meaning a space of a certain length between clock word bursts. Change with LEFT TOGGLE: display will change from extreme left: 145 ... 3 > AUTO > 2 ... 192 with numbers to the left of AUTO denoting "RESET AFTER SPACE" or ") < xxx > (" mode and the number of samples to reset word length upon, and numbers to the right "MANUAL mode" or "BITS=XXX" and the number of bits to reset on.

The first type ("BITS=XXX) sets the word length manually and will not reset on spaces after the trigger point but simply continuously counts bits, resetting on the selected word size. Use for example with a continuous stream of clock pulses without spaces between words. Bit numbering will simply jump across spaces if the setting does not correspond to the number of clock pulses in a burst and continue increasing. Also use this mode with bits set at 7 or 8 bits to display ASCII characters in chart.

"AUTO" resets word size on any space between bursts that is greater than 2 clock pulses. Can be used for virtually all transmissions, specially useful when word size varies.

The last case ") < xxx > (" manually sets the length of a space, in samples, that the word length will reset upon. There are 30 samples to a screen division, 6 to a minor division. This needs to be set to a value greater than the distance between clock pulses, but less than the space between the word clock bursts.

Useful when the word length changes during a transmission. It can also be used to manually synchronize the decoder to a continuous data stream that lacks the necessary space before the waveform trigger point to reset the word size, provided SOME space greater than a clock space (1/2 period) is available.

Default space before triggering is 60 samples or 2 divisions, regardless of the X position. Anything prior or to the left of this is disregarded.

Calibration - Wildcat 6.6

SECTION 10 / REPLACES THE LEGACY 6.5b STEPS

The 6.6 wizard calibrates Normal channels A and B together and, when enabled, the interleaved B-to-A path used by 144 MS/s. This supersedes the previous one-channel-at-a-time procedure. The original hardware operating-point adjustment (ADCoffset) is a separate setting and should not be confused with the ZERO and DC reference captures.

Preparation

- Use stable, low-noise DC reference and proper leads to both CH A and CH B. Avoid creating a ground loop with a noisy USB charger.
- The first phase is a clock-frequency adjustment. Provide a frequency counter if accurate time calibration is required.
- For the ZERO phase, physically ground both analogue inputs; do not apply the DC reference yet.
- For DC REF, apply the same known DC voltage to both CH A and CH B. Observe correct polarity and safe range.

Wizard sequence

Stage	What it does
CLOCK	1 MHz generator output; adjust frequency correction in ppm, K2 continues.
ZERO	Eight ranges, Normal A/B zero and four-point DAC checks; optional 144 B-to-A zero checks.
DC REF	Known voltage per range, gain and consistency checks; optional 144 measured reference/relative gain.
SAVE	Only after all eight ranges pass. K2 stores BATTERY profile; K3 stores AC ADAPTER profile.

Clock stage

The wizard outputs a nominal 1.000000 MHz square wave. Adjust the ppm value with the left rocker based on an external frequency counter. The supported adjustment is -999 to +999 ppm. Press K2 to continue or K4 to cancel without saving.

Key difference. Unlike legacy 6.5b, the modern ZERO step is shared across both Normal ADCs. B144 calibration is a separate measurement of the B ADC on CH A, not a duplicate independently calibrated analogue B input.

ZERO - Normal A/B and 144

SECTION 10 / CALIBRATION

Run ZERO

- Disconnect the reference and ground CH A and CH B. Press K2 to start ZERO; the wizard proceeds through all eight ranges.
- Normal path: it collects stable, valid blocks from both ADCs, then solves offset/DAC corrections with repeated checks.
- 144 path, when enabled: it routes the B ADC to analogue CH A and performs separate B-to-A zero and DAC response verification over the shared analogue A path.
- Do not remove grounding or move probes while zero/offset captures run; if a stage fails, use the on-screen retry instructions.

Measured capture quality

Guard	Current 6.6 implementation
Capture block	4096 paired ADC samples; one initial discard record.
Block-window stability	Three good blocks; accepted A/B spread up to 2 ADC codes, with extra monotonic-drift guard.
Intra-block time slices	Four slices; spread limit of 3 ADC codes.
Distribution filter	Median/MAD histogram estimates, bounded rail rejection, asymmetric-bias checks.
Overall capture deadline	2 s per capture attempt.

These figures explain the present acceptance logic. They are not suggested settings or user-editable calibration tolerances.

Ground loop and supply sensitivity

A charger can inject additional noise through the signal ground or power supply. A correct-looking averaged DC value alone does not guarantee the block passed the noise/distribution checks. Repeat calibration on a clean source if error 12 occurs; do not increase tolerances blindly.

144 terminology. ADC B is re-used on analogue CH A during interlaced operation. ZERO B144 measures the physical ADC and its route; the corrected B samples are later normalized against CH A.

DC REF, progress and saving

SECTION 10 / CALIBRATION

Apply DC reference

- Connect a stable, known positive DC voltage to both CH A and CH B. The same reference is required by the Normal A/B measurements.
- The top line displays the currently entered reference. Use the left rocker to adjust this value to the external measured voltage, not to the uncalibrated preview.
- Use the right rocker to select another of the eight ranges if needed. Press K2 to verify the selected range.
- Normal path: calculates the gain of A and B from freshly captured ZERO and reference readings, then checks DAC points.
- 144 path: performs an independent B-to-A reference verification using the shared A input and stores paired interlace coefficients.

Preview behavior in this code revision

The idle DC REF display periodically shows live voltage estimates for A and B. The preview is display-only and does not replace the fresh measurements taken on K2. The current preview accepts a block only when both ADC estimates pass. If one ADC becomes noisy, after about 1.5 s both live cells may show --. Independent per-channel preview validity is a proposed change, not yet present in commit 4df9eba.

Save after all eight ranges

Key	Saved profile / notes
K2	BATTERY (low-voltage calibration profile)
K3	AC ADAPTER (high-voltage calibration profile)
K4	Cancel/exit without saving

The 6.6 firmware uses redundant device-serial-number files XXXXXXXX.WPT and XXXXXXXX.WPB (1536 bytes per file). Each save targets the alternate generation, verifies it, and preserves the last valid calibration when a write fails. The Native/144 calibration information is kept together in one selected generation.

Power stability check - currently implemented

The ADC3 battery reading must remain within 2800-5200 mV and within +/-200 mV of the value captured at the start of ZERO. This reference is still used during subsequent DC REF and SAVE. Separate ZERO/DC REF starting baselines and filtered confirmation have been specified for future work but are not yet merged in the verified commit.

Calibration errors and diagnostics

SECTION 10 / CALIBRATION

Error reference

Code	Meaning / response
2	FIFO or capture setup failed. Recheck mode, hardware and retry.
3	ADC clipping/rails or centre outside accepted 10..245-code range. Check reference, range and offset.
5 / 11	Offset solution cannot fit or converge/stabilize. Check inputs and supply; inspect offset diagnostic panel.
6	DC gain response too small or gain outside accepted coefficient interval. Verify applied voltage and range.
7	144 relative gain/span invalid. Check B-to-A routing, calibration source and ADC headroom.
10	WPT/WPB save failed; previous RAM calibration is restored. Verify storage.
12	DC blocks did not settle before deadline (noise, distribution, time stability or insufficient valid blocks).
13	Four-point DAC response/return/slopes did not agree. Inspect displayed path, channel and residual.
14	Supply outside 2800..5200 mV or drift beyond 200 mV from current ZERO baseline.
15	No common DAC/ADC headroom for four calibration points.

When a charger produces errors

- Test on the battery with clean reference/source ground and note whether error 12 disappears.
- If error 14 appears on battery, observe battery voltage, load and drift over the full wizard; this code does not yet separately re-baseline DC REF.
- Use the RAW CAPTURE DIAGNOSTICS panel to compare A/B means, spread, clips and raw extrema. A failure may be one ADC only even if preview suppresses both values.

Do not bypass integrity checks

A failed calibration must not commit partial new coefficients. Retry the failed step after correcting the input rather than assuming a preview-looking voltage proves a valid result.

Hardware and firmware compatibility

SECTION 11 / UPDATED FOR 6.6

Supported combinations

Component	Guidance
DS203 hardware	Wildcat 6.6 in this project targets HW 2.81 and 2.82. Older 2.6-2.72 use their corresponding legacy firmware.
System	Project README lists SYS 1.64 + AlterBIOS 0.7; it remains separate from the application.
FPGA W1.1	Supports compatible Normal acquisition/oversampling paths, but not the validated W1.2-only 144 UI.
FPGA W1.2	Required for 144 MS/s; the firmware checks the image identity and CRC before entry.
Application	Wildcat V6.6 uses multiple disjoint APP1A/B/C flash regions, not one flat contiguous image.

Application flash regions

Region	Address / limit	Capacity
APP1A	0x0800C000 - 0x0801BFFF	64 KiB
APP1B	0x0802C000 - 0x0803FFFF	80 KiB
APP1C	0x08047000 - 0x080573FF	65 KiB
FPGA W1.2	0x0805C800 - 0x0807D7B7	135,096 B

These addresses are taken from the current linker/export verification scripts and FPGA CRC check. They describe distinct programmed regions. Firmware update tools must not flatten all files into one uninterrupted binary image.

Basic flash safety

- Backup original firmware and settings before programming.
- Flash only the images meant for this hardware revision; do not overwrite SYS, FPGA or reserved space with an application image.
- For updates, use the generated ADR/BIN region pairs rather than guessing a single address from the HEX length.

Release notes and known limits

SECTION 11 / VERIFIED CODE STATE

What 6.6 adds beyond the illustrated 6.5b guide

- 144 MS/s two-ADC interleaved acquisition of CH A on FPGA W1.2, with corrected A/B phase and recorded timing.
- Dedicated 144 controls, three fast timebases, digital C/D trigger support, ADC rail indicators, analog/digital metering, cursors and 512-point FFT.
- Calibrated interlace B-to-A relative path tied to the Normal A reference, including eight-range ZERO/DC REF and four-point DAC checks.
- Redundant, versioned WPT/WPB Native+144 calibration profiles with verification, rollback and guarded CFG updates.
- Improved median/MAD capture-quality checks, zero/reference diagnostics and common battery/adapter profile saves.

Important differences from a planned future revision

Feature discussed	Status at commit 4df9eba
Separate validity/timeouts for A/B DC REF preview	Not present; either invalid block suppresses both live readings.
Separate supply baseline for ZERO and DC REF	Not present; +/-200 mV is currently measured against ZERO start.
Confirmed/filtered battery ADC3 changes	Not present as a new independent-sample confirmation algorithm.
Small-buffer or 144 LIVE high-frame-rate mode	Not present. 144 uses the 4096-word record.
Global flash-size/refactoring cleanup	Not part of this documented build.

Revision scope

The original illustrated operating guide and hardware figures remain the source for long-standing Normal, chart, generator, serial decoding and classic FFT functions. Sections 04A, 10 and 11 of this edition supersede older statements about 144 acquisition and calibration. Code verified from bales0/DSO203-HW-2.81, commit 4df9eba (21 October 2026).

SECTION 12

Credits and sources

I wish to extend appreciation to the SEEED forum developers of the program this version is based on: marcosin pmoss69 JackTheVendicator As well as to the original authors for making the source available

And also to bobtidey for the support code for 8Mb devices JPA for the Alterbios patch. Jerson for suggesting a beep to indicate long button press. (V4 includes a visual indicator for this as well)

Source: "Wildcat Revision 5.6 / 6.5 User Guide" (21 pages), using the sections applicable to Wildcat 6.5b on HW 2.81 and 2.82. Screen captures and function identification: project README at github.com/bales0/DS203-DSOQuad. Older revision history is retained where needed for technical context.

EDITORIAL NOTE - 6.6 EDITION

Original 5.6/6.5 credits and artwork are preserved. This edition also uses the current Wildcat 6.6 source from github.com/bales0/DSO203-HW-2.81 (commit 4df9eba, 21 October 2026). The legacy repository address below is retained only as historical provenance.